

Olfactory ensheathing cells from adult female rats are hybrid glia that promote neural repair

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eLife Assessment

This study unveils **important** data describing cell states of olfactory ensheathing cells, and how these cell states may relate to repair after spinal cord injury. The framework used for characterizing these cells is **solid**, although the study would be strengthened by additional quantification of immunohistochemical data and by complementing expression data with functional outcomes. This work will be of interest to stem cell biologists and spinal cord injury researchers.

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Abstract

Olfactory ensheathing cells (OECs) are unique glial cells found in both central and peripheral nervous systems where they support continuous axonal outgrowth of olfactory sensory neurons to their targets. Previously we reported that following severe spinal cord injury, OECs transplanted near the injury site modify the inhibitory glial scar and facilitate axon regeneration past the scar border and into the lesion. To better understand the mechanisms underlying the reparative properties of OECs, we used single-cell RNA-sequencing of OECs from adult rats to study their gene expression programs. Our analyses revealed five diverse OEC subtypes, each expressing novel marker genes and pathways indicative of progenitor, axonal regeneration, secreted molecules, or microglia-like functions. We found substantial overlap of OEC genes with those of Schwann cells, but also with microglia, astrocytes, and oligodendrocytes. We confirmed established markers on cultured OECs, and localized select top genes of OEC subtypes in olfactory bulb tissue. We also show that OECs secrete Reelin and Connective tissue growth factor, extracellular matrix molecules which are important for neural repair and axonal outgrowth. Our results support that OECs are a unique hybrid glia, some with progenitor characteristics, and that their gene expression patterns indicate functions related to wound healing, injury repair and axonal regeneration.

Introduction

Olfactory sensory neurons (OSNs) are remarkable as they undergo life-long neurogenesis within the adult olfactory epithelium (Graziadei & Monti Graziadei, 1985 [DOI](#)), and their axons are exclusively associated with a specialized type of glia, olfactory ensheathing cells (OECs; Doucette, 1990 [DOI](#)). OECs provide the outgrowth and guidance factors needed for OSN axons to reach their glomerular targets and exhibit regenerative abilities in response to injury (Doucette, 1991 [DOI](#); Li et al., 2005 [DOI](#); Williams et al., 2004 [DOI](#)). OECs have received considerable interest due to their ability to support axonal outgrowth even after severe or complete spinal cord transection in rodents, dogs and humans (Granger et al., 2012 [DOI](#); Khankan et al., 2016 [DOI](#); Ramon-Cueto et al., 2000 [DOI](#); Tabakow et al., 2014 [DOI](#); Takeoka et al., 2011 [DOI](#); Thornton et al., 2018 [DOI](#)).

The regenerative capacity of adult spinal cord neurons is inhibited by the lesion site environment that includes a reactive glial scar, together with invading meningeal fibroblasts and multiple immune cells (Burda & Sofroniew, 2014 [DOI](#); Wanner et al., 2013 [DOI](#)). The lesion core is composed of non-neural tissue that inhibits axonal outgrowth due to the upregulation and secretion of chondroitin sulfate proteoglycans that impede axon regeneration (Cregg et al., 2014 [DOI](#)). Some studies of severe or complete spinal cord transection in rats followed by the transplantation of OECs provided evidence that OECs surround axon bundles which regenerate into and occasionally across the injury site (Khankan et al., 2016 [DOI](#); Ramon-Cueto et al., 2000 [DOI](#); Takeoka et al., 2011 [DOI](#); Thornton et al., 2018 [DOI](#)). In combination with *in vitro* studies, detailed analyses of injury sites suggest that OECs: 1) function as phagocytes to clear degenerating axonal debris and necrotic cells (Khankan et al., 2016 [DOI](#); Nazareth et al., 2020 [DOI](#); Su et al., 2013 [DOI](#)), 2) modulate the immune response to injury (Khankan et al., 2016 [DOI](#); Vincent et al., 2007 [DOI](#)), and 3) interact favorably with the glial scar and lesion core that form after injury *in vitro*, (Lakatos et al., 2003 [DOI](#); Lakatos et al., 2000 [DOI](#)) and *in vivo*, (Khankan et al., 2016 [DOI](#); Thornton et al., 2018 [DOI](#)). OEC transplantation following complete spinal cord transection creates an environment that is conducive to neural repair (Khankan et al., 2016 [DOI](#); Takeoka et al., 2011 [DOI](#); Thornton et al., 2018 [DOI](#)). OECs secrete molecules that stimulate axon outgrowth such as Brain-derived neurotrophic factor (Bdnf), Nerve growth factor (Ngf), and Laminin (Ruitenberg et al., 2003 [DOI](#); Runyan & Phelps, 2009 [DOI](#); Woodhall et al., 2001 [DOI](#)) and enhance cell-to-cell mediated interactions between neurites and OECs in an inhibitory environment (Chung et al., 2004 [DOI](#); Khankan et al., 2015 [DOI](#); Windus et al., 2007 [DOI](#)). Khankan et al. (2015) [DOI](#) showed that neurites grew 2-3 times longer in growth-inhibitory areas of multicellular ‘scar-like’ cultures when they were directly associated with olfactory bulb-derived OECs (OB-OECs), but the molecular mechanisms are unknown.

The outcomes of OEC transplantation studies after spinal cord injury vary substantially in the literature due to many technical differences in their experimental designs. The source of OECs has a great impact on the outcome, with OB-OECs showing more promise than lamina propria-derived OECs, and purified, freshly-prepared OECs being required for optimal OEC survival. Other important variables include the severity of the injury (hemisection to complete spinal cord transection), the age of the spinal cord injured host (early postnatal versus adult), and OEC transplant strategies (delayed or acute transplantation, cell transplants with or without a matrix; Franssen et al., 2007 [DOI](#)). Franssen et al. (2007) [DOI](#) evaluated studies that used only OECs as a transplant, and reported that 41 out of 56 studies showed positive effects, such as OEC stimulation of regeneration, positive interactions with the glial scar and remyelination of axons. More recent systematic reviews and meta-analyses on the effects of OEC transplantation following different spinal cord injury models reported that OECs significantly improved locomotor function (Watzlawick et al., 2016 [DOI](#); Nakjavan-Shahraki et al., 2018), but did not improve neuropathic pain (Nakjavan-Shahraki et al., 2018).

The question addressed in this study is how OB-OECs perform the diverse functions associated with neural repair. We hypothesize that there exist OB-OEC subtypes, and that their respective gene programs and secreted molecules underlie their multifaceted reparative activities. Here we identify the OB-OEC subtypes and their molecular programs that contribute to injury repair, such as the growth-stimulating secreted and cell adhesion molecules involved in OB-OEC interactions that promote neurite and axonal outgrowth. We used single-cell RNA-sequencing (scRNA-seq) to characterize the immunopurified female rat OB-OECs (hereafter called OECs unless stated otherwise) that are similar to those transplanted into our previous female rat spinal cord injury studies. We first compared the genes expressed by purified OECs with those expressed by the ‘leftover cells’, i.e., the cells which are not selected by the panning procedure, to determine if the purification process selected for specific OEC subtypes. After confirming the characteristic OEC markers, our scRNA-seq data revealed five OEC subtypes which expressed unique marker genes and were further characterized and confirmed experimentally. Finally, we examined interesting extracellular matrix (ECM) molecules secreted by OECs, Reelin (*Reln*) and Connective tissue growth factor (*Ccn2/Ctgf*), both of which may facilitate axonal outgrowth into the inhibitory injury core environment.

Results

One particularly challenging aspect facing neural repair of spinal cord injury (SCI) is to facilitate axonal outgrowth and eventually regeneration across the inhibitory injury site. A common finding in our studies testing the effects of OEC transplantation following SCI is that OECs modify the injury site so that some axons project into the lesion core surrounded by OECs (Dixie, 2019; Khankan et al., 2016; Takeoka et al., 2011; Thornton et al., 2018). **Figure 1** shows an enlargement of an injury site which contains green fluorescent protein (GFP)-labeled OECs in the glial fibrillary acidic protein (GFAP)-negative injury core between the two GFAP-positive spinal cord stumps. The OECs in the injury core are associated with a few serotonergic-(**Figure 1a-f**) and many neurofilament-labeled axons (**Figure 1g-l**). We note, however, that such bridge formation is rare following severe spinal cord injury in adult mammals. To learn more about possible mechanisms by which OECs support axon regeneration, we prepared immunopurified OECs and examined their gene expression and diversity with scRNA-seq.

Unbiased scRNA-seq of immunopurified OECs and ‘leftover’ controls distinguishes OECs, microglia, and fibroblasts

Using scRNA-seq, we sequenced a total of 65,481 cells across 7 samples (n=3 OEC preparations and 4 ‘leftover’ controls) that passed quality control, with an average of 9,354 cells per sample. Cell clusters were visualized using t-distributed stochastic neighbor embedding (tSNE). The tSNE plot showed that cells from immunopanned OEC samples were separated from the cells from the leftover samples (**Figure 2a**). Cell clustering analysis defined a total of seven clusters (**Figure 2b**), each of which showed expression patterns of marker genes that distinguish one cluster from the others (**Figure 2c**). Based on previously reported cell type marker genes for fibroblasts and major glial cell types including OECs, astrocytes, oligodendrocytes, and microglia, we found elevated expression of OEC marker genes in clusters 2, 3 and 7, microglia marker genes in clusters 4, 6, and 7, and fibroblast marker genes in clusters 0, 1, and 5 (**Figure 2d**, Suppl. Figure 1). In contrast, except a few select genes, the majority of markers for astrocytes and oligodendrocytes showed low expression (**Figure 2d**). The cell cluster tSNE plot with cell type labels in **Figure 2e**, and the distinct expression patterns of cell type markers are confirmed in a dot plot in **Figure 2f**. As expected, many cells enriched in the leftover controls were defined as fibroblasts and microglia based on the corresponding known marker genes, whereas cells from the immunopanned OEC samples showed high expression of OEC marker genes (**Figure 2a** vs. 2e). After cell type assignment, a dot plot clearly depicts distinct expression of cell type-specific marker

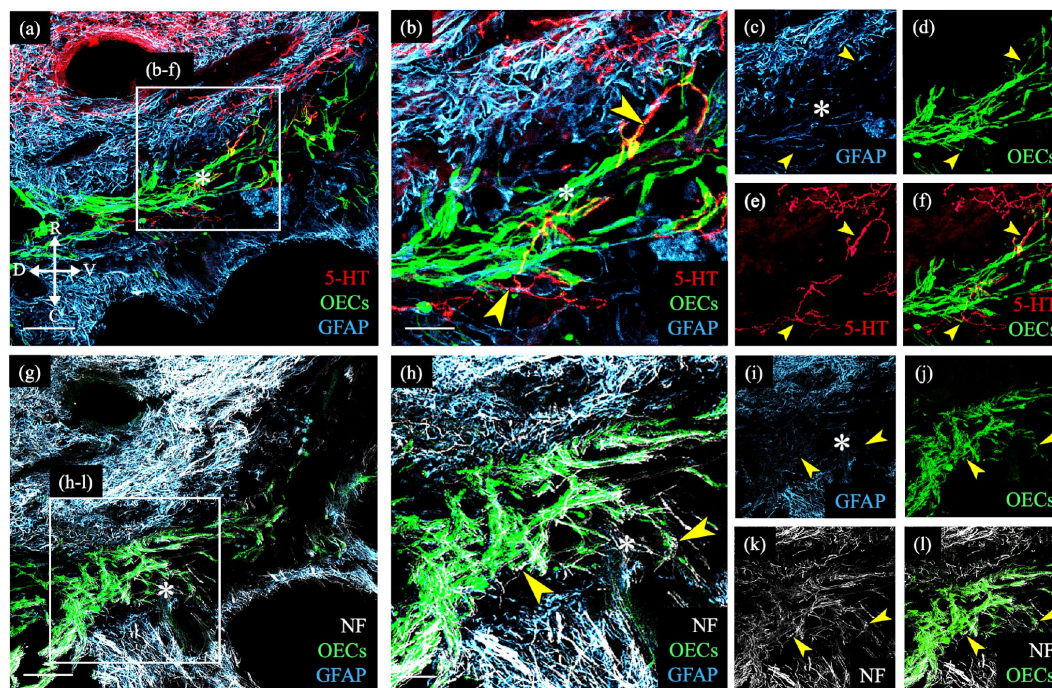


Figure 1.

Transplanted GFP-OECs in the center of the lesion core associate with numerous axons.

Sagittal sections show rostral and caudal glial scar borders of the injury site which are identified with glial fibrillary acidic protein (GFAP, blue). The GFAP-negative lesion core contains GFP-OECs (green) and is marked with asterisks. Arrowheads mark axons crossing into the lesion core. **(a-f)** Serotonergic axons (5-HT, red) are found in the rostral spinal cord stump and associate with OECs (green) in the lesion core. Single channels for GFAP **(c)**, OECs **(d)**, 5-HT **(e)**, and a combination of 5-HT and OECs **(f)**. **(g-l)** Nearby injury site section from the same rat **(a)**. Numerous neurofilament-positive axons (white) are associated with the OECs (green) in the lesion core. Single channels for GFAP **(i)**, OECs **(j)**, neurofilament **(k)**, and a combination of neurofilament and OECs **(l)**. Scale bars: a, g = 500 μ m; b, h = 100 μ m. Reprinted from: Dixie, KL, 2019, *UCLA*. ProQuest ID: Dixie_ucla_0031D_18445.

genes (**Figure 2f** [↗](#)). Our scRNA-seq results support the expected cell type composition following the immunopanning procedure. All marker genes for each major cell type, including statistics, are in Suppl. Table 1.

We next asked if there were differences between cultures of immunopurified OECs and OECs grown in ‘leftover’ cultures. While our previous spinal cord injury studies transplanted 90-95% purified OECs, other labs culture and implant less purified OECs and thus many of the neighboring cells, i.e., primarily fibroblasts and microglia, would be included. Our comparison showed that purified OECs express higher levels of *Cryab* (stress protection, inflammation inhibition), *Nqo1* (antioxidant protection, stress adaptation), and *Postn* (cell proliferation, survival, migration) genes (**Figure 2g** [↗](#)), whereas OECs in leftover cultures with fibroblasts and microglia express higher levels of *Apoe* (lipid transport and glial function), *Calcb* (inflammation and pain modulation), and *Mgp* (ECM calcium inhibitor; **Figure 2g** [↗](#)) than purified OECs. These results indicate immunopurified OECs may have better stress response and proliferative and survival capacity than the controls.

Immunofluorescence verification of typical OEC marker genes revealed by scRNA-seq

To validate the OEC markers revealed by scRNA-seq, we re-cultured extra cells not used for sequencing for immunocytochemical confirmation. Because our adult OECs are immunopurified with anti-nerve growth factor receptor p75 (Ngfr^{p75}), we expected and found OEC samples heavily enriched in Ngfr^{p75} in both immunofluorescence (**Figure 3a** [↗](#)) and scRNA-seq (**Figure 3b** [↗](#)). Next, we detected the high expression of two common glial markers, Blbp (*Fabp7*; **Figure 3c** [↗](#), **3d** [↗](#)) and S100β (**Figure 3e** [↗](#), **3f** [↗](#)), in OECs. Sox10 is expressed in the nuclei of neural crest-derived cells such as OECs and Schwann cells, and our cultures showed high expression levels in scRNA-seq and immunocytochemistry (**Figure 3c** [↗](#), **3g** [↗](#), **3k** [↗](#), **3m** [↗](#)). Three cell adhesion molecules, L1-Cam (**Figure 3g** [↗](#), **3h** [↗](#); Runyan & Phelps, 2009 [↗](#); Witheford et al., 2013 [↗](#)), N-Cam (**Figure 3i** [↗](#), **3j** [↗](#)), and N-Cadherins (**Figure 3k** [↗](#), **3l** [↗](#)), are also expressed in purified OEC cultures. In addition, other previously reported OEC genes are now verified by scRNA-seq, including the pro-myelinating Ciliary neurotrophic factor (*Cntf*, **Figure 3n** [↗](#); Asan et al., 2003 [↗](#); Roet et al., 2011 [↗](#)), the laminin receptor α7 integrin (*Itga7*, **Figure 3o** [↗](#); Ingram et al., 2016 [↗](#)), and the myelin related gene P0 (*Mpz*, **Figure 3p** [↗](#); Sasaki et al., 2006 [↗](#)). Thus, well-characterized OEC markers were detected by scRNA-seq and most are widely distributed among OEC clusters.

In addition to confirming select genes, we also compared the 209 significant OEC markers identified from our scRNA-seq data with the 309 genes from the meta-analysis of five OEC microarray studies on cultured early-passage OB-OECs versus other tissues (Roet et al., 2011 [↗](#)). We found 63 genes overlapping between our OEC markers and the published markers, representing a 15.3-fold overlap enrichment (p-value: 4.3 e-58), further supporting the reliability of our scRNA-seq findings.

Subclustering analysis of OECs revealed refined OEC subtypes and their top marker genes

Next, we extracted OECs and performed a subclustering analysis with OECs alone, and found five distinct subclusters (**Figure 4a** [↗](#)). All subclusters except for cluster 3 expressed previously identified markers of OECs as well as subsets of marker genes for astrocytes, oligodendrocytes, and Schwann cells (**Figures 4b, c** [↗](#)). The percentage of OECs within each cluster that express a gene, the percentage of OECs in all other subtypes that express a gene and their p values are found in Suppl. Table 2. Interestingly, cluster 3 showed expression of both microglia and OEC markers. The top 20 genes expressed by each OEC subtype were investigated to evaluate potential

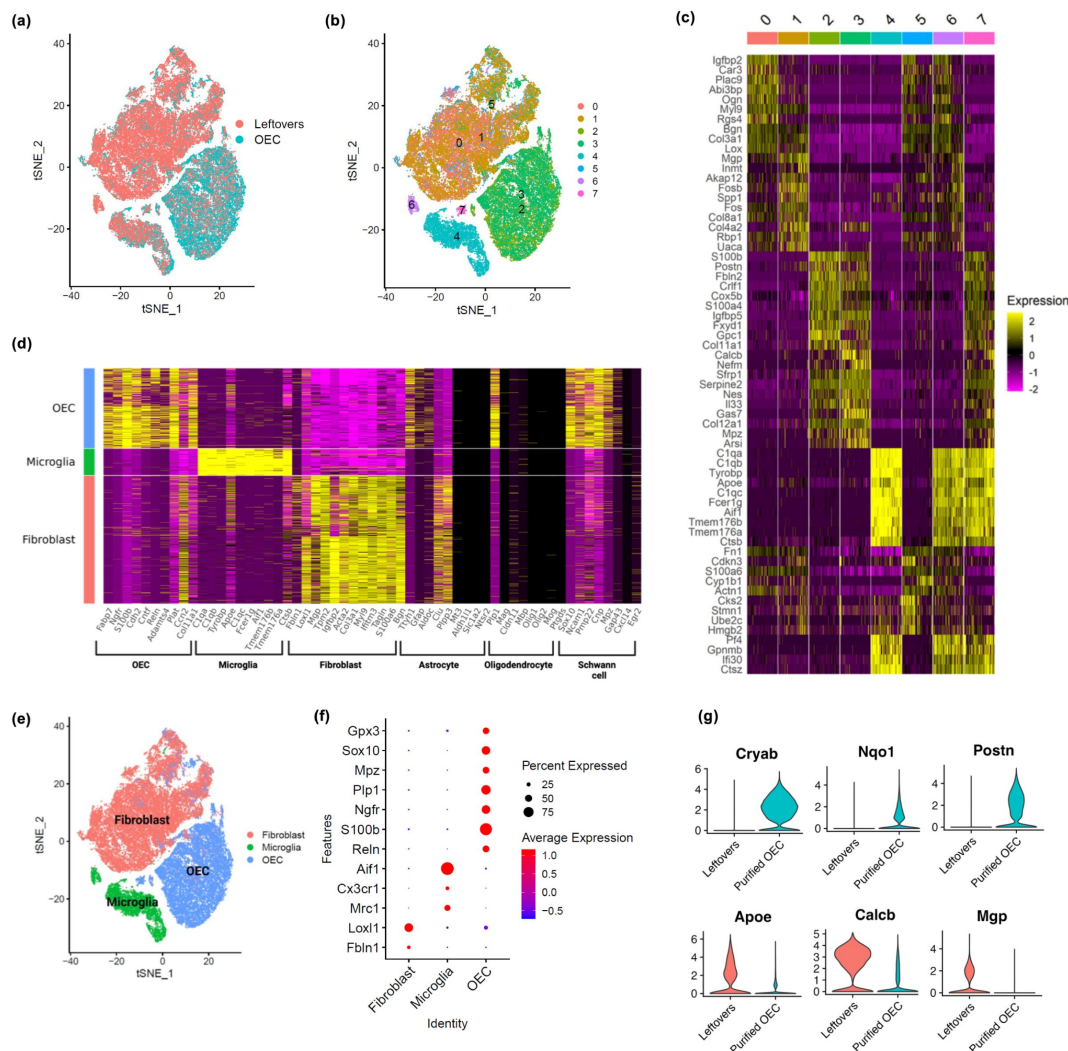


Figure 2.

scRNA-seq results show distinct clusters of OEC and leftover cell samples.

(a) Cells in tSNE plot colored by sample source, with cells from leftover samples in pink and cells from immunopurified OECs in cyan. (b) Clustering analysis revealed a total of 7 distinct cell clusters, each indicated with a different color. (c) A heatmap showing expression patterns of top marker genes of individual cell clusters. (d) Cell clusters showed distinct expression patterns for known cell type markers for fibroblasts, microglia, and OECs. Clusters 0, 1, and 5 had high expression of fibroblast markers and are labeled as fibroblast in the Y-axis; clusters 4, 6, and 7 showed high expression of microglia markers and are labeled as microglia; clusters 2, 3, and 7 showed high expression of OEC markers and are labeled as OEC. Known marker genes for different cell types are on the X-axis. Feature plots and violin plots for select marker genes are in Suppl. Figure 1. (e) Based on known cell type markers, cell clusters in tSNE plot were labeled with the corresponding cell types. (f) The expression of cell-type specific marker genes is depicted in a dot plot. (g) Genes that distinguish purified OECs vs OECs in leftover cultures are shown. The top 3 genes were higher in purified OECs, whereas the bottom 3 genes were higher in 'leftover' cultures.

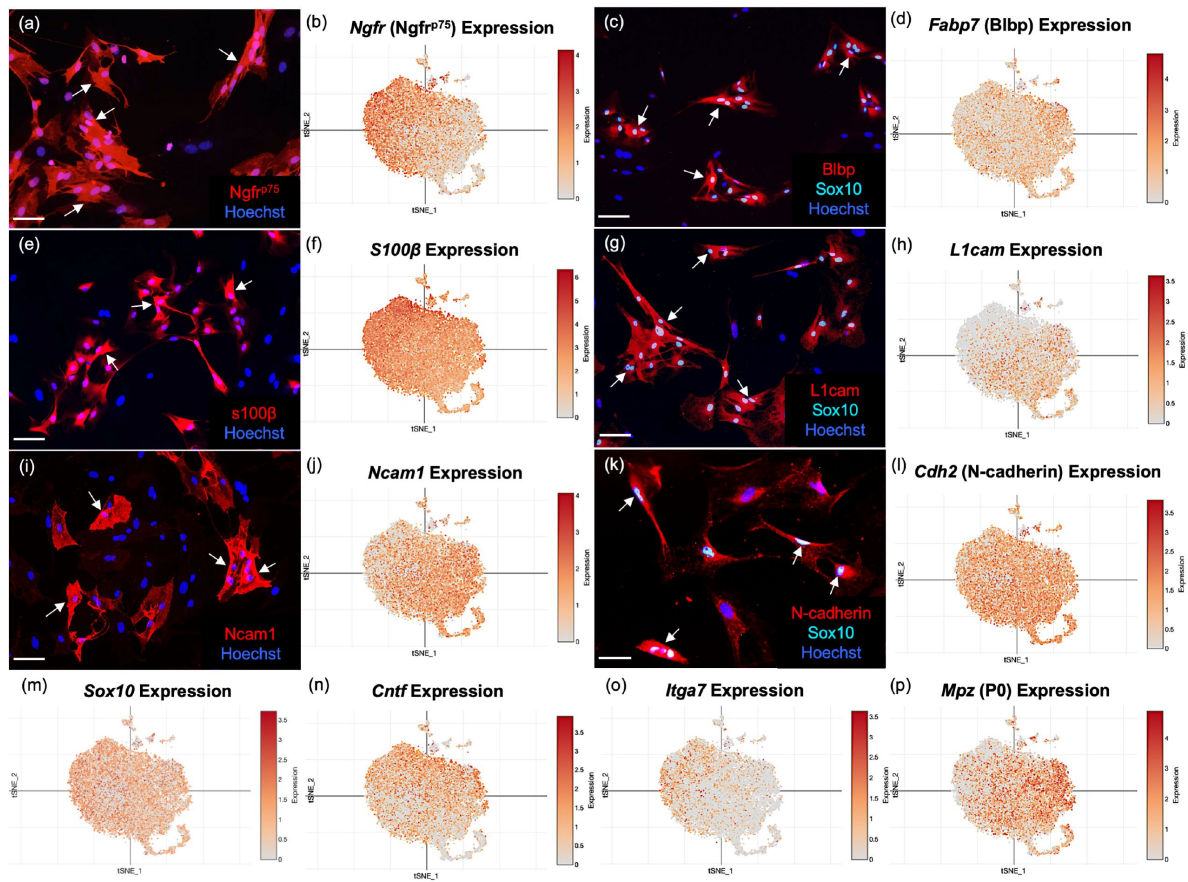


Figure 3.

Well-established OEC markers are revealed by scRNA-seq and immunofluorescence.

OEC cultures were replated from leftover cells prepared for scRNA-seq. Immunolabeled OECs are marked with arrows and all cell nuclei are stained with Hoechst (blue nuclei). tSNE maps of the gene expression in the 5 clusters are shown next to the protein expression in a-l. **(a, b)** Cultured OECs express Ngfr^{p75} protein and Ngfr^{p75} gene expression. **(c, d)** Blbp and Sox10 immunoreactive OECs with Fabp7 gene expression. **(e, f)** S100 β -labeled OECs together with $\text{S100}\beta$ expression. **(g, h)** L1cam and Sox 10 labeling next to L1cam expression. **(i, j)** Ncam1 protein and gene expression. **(k, l)** N-Cadherin and Sox10 markers with Cdh2 expression. **(m-p)** scRNA-seq data for Sox10 , Cntf , Itga7 , and Mpz (references in text). Scale bars: a, c, e, g, i, k = 50 μm .

functional differences between the subtypes (**Figure 4c** [↗](#); Suppl. Table 3), and their enriched pathways (**Figure 4d** [↗](#)). Most references for the specific genes within the clusters are found in Suppl. Table 3.

Cluster 0 is rich in matricellular proteins

Pathway enrichment analysis showed that ECM organization and collagen biosynthesis were significantly enriched in the marker genes for the largest OEC subcluster 0 (**Figure 4d** [↗](#)). Two of the top 10 genes are members of the Ccn family of matricellular proteins, *Ccn2/Ctgf* (Connective tissue growth factor; **Figure 5a** [↗](#)) and *Ccn3/NOV* (nephroblastoma overexpressed gene; **Figure 5b** [↗](#), Suppl. Table 3). Both are secreted ECM proteins which regulate the activity of growth factors and cytokines, function in wound healing, cell adhesion, and injury repair. The presence of *Ccn3* in OECs is novel, while *Ccn2/Ctgf* is well established (Lamond & Barnett, 2013 [↗](#); Roet et al., 2011 [↗](#)). In **figures 5c** [↗](#), c1-2, we show that *Ccn2* and *Sox10* are highly expressed in our cultured OECs. Because Mokalled et al. (2016) [↗](#) reported that *Ctgfa* is a critical factor required for spontaneous axon regeneration following spinal cord injury site in zebrafish, we asked if GFP-labeled OECs transplanted 2 weeks following a complete spinal cord transection in adult rats, also expressed *Ccn2/Ctgf*. We found high levels of *Ctgf* expression in GFP-OECs that bridged much of the injury site and also detected *Ccn2* on near-by cells (**Figure 5d** [↗](#), d1-2). GFP-labeled fibroblast transplantations served as controls. Other top ECM genes are *Serpinf1* (Serine protease inhibitor), *Fbln2* (Fibulin2), *(Fibronectin-1), and *Col11a1* (Collagen, type XI, alpha 1; Suppl. Table 3).*

The top-ranked marker gene for cluster 0 is *Pmp22* (Peripheral myelin protein 22), a tetraspan membrane glycoprotein that is highly expressed in myelinating Schwann cells, and contributes to the membrane organization of compact peripheral myelin. The third-ranked gene, *Gldn* (Gliomedin) is secreted by Schwann cells and contributes to the formation of the nodes of Ranvier. *Fst* (Follistatin) also plays a role in myelination and is expressed in areas of adult neurogenesis.

Many of the top cluster 0 genes are found in other glial cells: *Fbln2* is secreted by astrocytes, *Ccn2/Ctgf* is expressed by astrocytes and Schwann cells, *Nqo1* (NAD(P)H dehydrogenase, quinone 1 enzyme) is found in Bergmann glia, astrocytes, and oligodendrocytes, and *Marcks* (Myristoylated alanine-rich C-kinase substrate protein) regulates radial glial function and is found in astrocytes and oligodendrocytes. In addition, *Cd200* is expressed by astrocytes and oligodendrocytes, and binds to its receptor, *Cd200R*, on microglial cells. Thus, the OECs in cluster 0 express high levels of genes found in other glia, many of which are related to ECM and myelination.

Cluster 1 contains classic OEC markers

Three of the top five genes in cluster 1, *Pcsk1* (proprotein convertase PC1), *Gal* (neuropeptide and member of the corticotropin-releasing factor family) and *Ucn2* (Urocortin-2) were reported in the meta-analysis by Roet et al. (2011) [↗](#). Galanin is expressed by neural progenitor cells, promotes neuronal differentiation in the subventricular zone and is involved in oligodendrocyte survival.

A well-known oligodendrocyte and Schwann gene associated with myelin, *Cnp* (2-3-cyclic nucleotide 3-phosphodiesterase), was high in OEC cluster 1 in addition to the cytokine *Il11* (Interleukin 11) that is secreted by astrocytes and enhances oligodendrocyte survival, maturation and myelin formation. The chondroitin sulfate proteoglycan Versican (*Vcan*) is another top gene expressed by astrocytes and oligodendrocyte progenitor cells.

Many of the genes and pathways that characterize cluster 1 are involved in nervous system development and axon regeneration (*Atf3*, *Btc*, *Gap43*, *Ngfr* ^{p75}, *Bdnf*, *Pcsk1*), and are found in other glia (Suppl. Table 3). Interestingly, after nerve injury, *Atf3* (cyclic AMP-dependent transcription factor 3) is upregulated by Schwann cells in the degenerating distal nerve stump and downregulated after axon regeneration is complete (Hunt et al., 2004 [↗](#)). *Btc* (Betacellulin), part of the Epidermal growth factor (Egf) family and a ligand for *Egfr*, also is expressed by Schwann cells

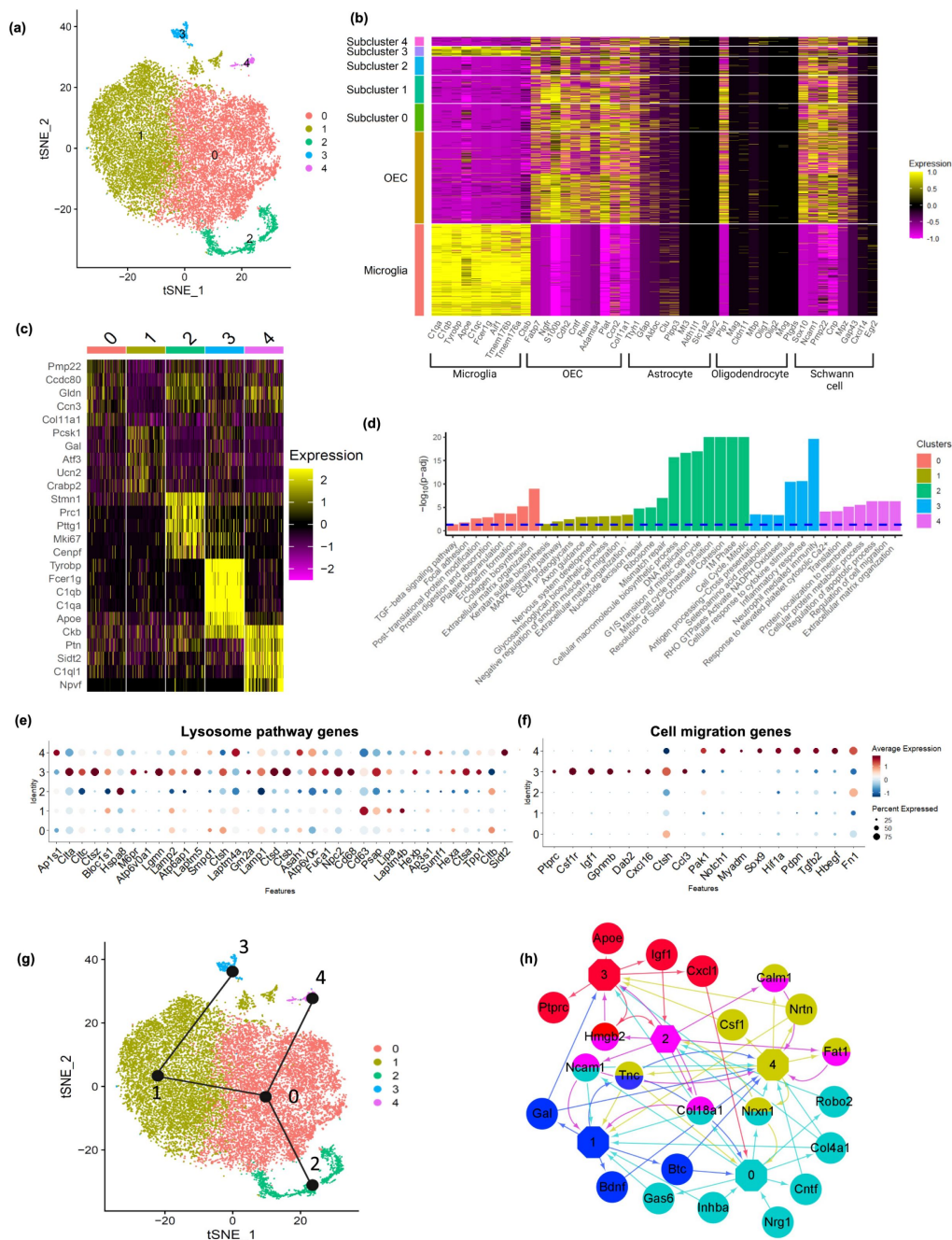


Figure 4.

OEC subclusters, marker genes, and enriched pathways.

(a) Clustering analysis of OECs revealed five separate clusters (0-4). **(b)** Heatmap depicts the expression patterns of known marker genes of glial cell types (x-axis) versus microglia, all OECs and OEC subclusters (y-axis). OEC subclusters express select markers of other glial cell types. **(c)** Heatmap depicts the top five marker genes (y-axis) for each OEC subcluster (x-axis). **(d)** Pathways associated with marker genes of different OEC clusters are shown. The dashed line indicates the false discovery rate (FDR) < 0.05 in the pathway analysis. **(e)** Dot plot showing that cluster 3 has higher potential lysosomal function based on lysosome pathway genes than the other clusters. **(f)** Both cluster 3 and 4 express select genes involved in positive regulation of cell migration. **(g)** Trajectory analysis reveals two trajectories, one including subclusters 2, 0, and 4, and another involving subclusters 2, 0, 1, and 3. **(h)** NicheNet ligand-receptor network plot demonstrates intracellular communication between OEC subtypes. Hexagons represent subclusters 0 to 4 and circles indicate ligands secreted from subclusters. Edges (arrows) point to clusters where receptors of the ligands are expressed. Different colors represent different OEC subclusters.

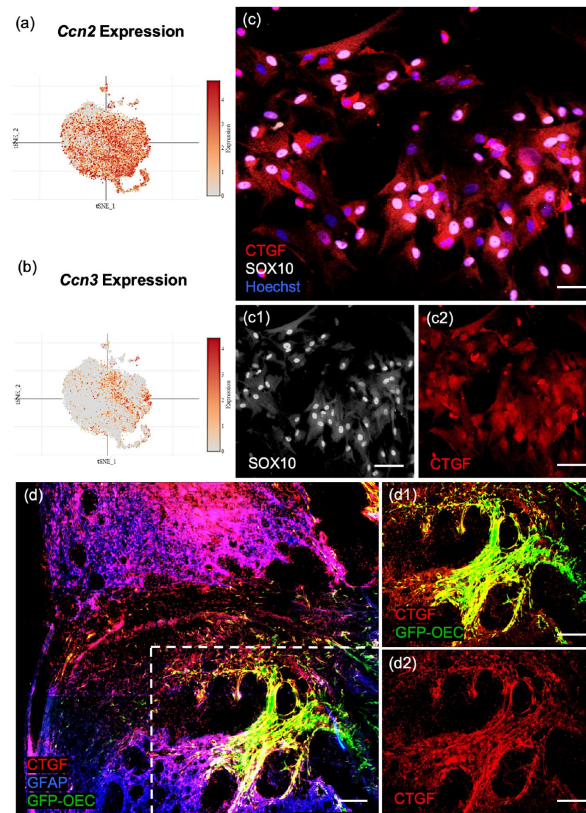


Figure 5.

Confirmation of *Ccn2/Ctgf* (Connective tissue growth factor) in cultured OECs and following OEC implantation after spinal cord injury.

(a, b) The scRNA-seq plots of the 9th (*Ccn2/Ctgf*) and 4th (*Ccn3/Nov*) highest-ranked marker genes are strongly expressed in subcluster 0. (c, c1, c2) The well-known matricellular protein Connective tissue growth factor (Ctgf) identifies OECs in cell cultures (red, c, c2), and Sox10 nuclear expression (white nuclei, c, c2) confirms they are neural crest-derived cells. (d, d1, d2) A sagittal section from a rat that received a complete spinal cord transection followed by Green Fluorescent Protein (GFP)-OEC implantation was fixed 2 weeks postinjury. Glial fibrillary acidic protein (d, Gfap, blue) marks the edges of the borders of the glial scar. GFP-OECs (green) that express Ctgf (red) in the injury site are outlined by the box in d. High expression of Ctgf is detected by GFP-OECs that bridge part of the injury site in d1. The single channel of Ctgf is shown in d2. Scale bars: c-c2 = 50 μ m, d-d2 = 250 μ m.

after nerve injury in a pattern similar to that of *Atf3*, as is Brain-derived neurotrophic factor (*Bdnf*). In addition, the proprotein convertase PC1 (*Pcsk1*) cleaves pro-Bdnf into its active form in Schwann cells and therefore also contributes to axon outgrowth. Together, the OECs in cluster 1 secrete a number of important growth factors associated with regeneration, including a novel one, *Atf3*, reported here (Suppl. Table 3).

Cluster 2 represents a distinctive proliferative OEC subtype

Most top genes in cluster 2 (**Figure 6a-f**) regulate the cell cycle (*Mki67*, *Stmn1*, *Cdk1*, *Cks2*, *Cdkn3*, *Cdca8*, *Cdca3*), are associated with mitosis (*Cenpf*, *Spc24*, *Tpx*, *Ckap2*, *Prc1*, *Ube2c*, *Top2a*), or contribute to DNA replication and repair (*Dut*, *Fam111a*, Suppl. Table 3). The top ranked gene, *Stmn1* (Stathmin 1; **Figure 6e**), is a microtubule destabilizing protein that is widely expressed in other OEC clusters and in areas of adult neurogenesis. Pathway enrichment confirms the proliferative property of cluster 2 with pathways such as G1/S transition of mitotic cell cycle and DNA replication (**Figure 4d**). These results show that there are distinct OEC progenitors in our cultures within the olfactory nerve layer of the adult olfactory bulb.

Mki67 is a well-known proliferation gene concentrated in cluster 2 (**Figure 6f**). Previous studies suggested the presence of two morphologically distinct OECs – the Schwann-cell-like, spindle-shaped OECs with high *Ngfr*^{p75} expression and the astrocyte-like, large flat OECs with low *Ngfr*^{p75} (Franceschini & Barnett, 1996). To determine if the proliferative OECs differ in appearance from adult OECs, and whether there is concordance between our OEC subtypes based on gene expression markers and previously described morphology-based OEC subtyping (Franceschini & Barnett, 1996), we analyzed OECs identified with the anti-Ki67 nuclear marker and anti-*Ngfr*^{p75} (**Figure 6g-h**). Of the Ki67-positive OECs in our cultures, 24% ± 8% were strongly *Ngfr*^{p75}-positive and spindle-shaped, whereas 76% ± 8% were flat and weakly *Ngfr*^{p75}-labeled (n=4 cultures, *p* = 0.023). Here we show that a large percentage (~3/4th) of proliferative OECs are characterized by large, flat morphology and weak *Ngfr*^{p75} expression resembling the previously described morphology-based astrocyte-like subtype. Our results indicate the two types of OEC classifications share certain degrees of overlap, indicating similarities but also differences between the two classification methods.

Cluster 3 resembles both microglia and OECs

In addition to the typical OEC markers (*Ngfr*^{p75}, *S100b*, and *Sox10*), cluster 3 also expressed numerous microglia markers (**Figure 4b**). In fact, all top 20 genes in cluster 3 are expressed in microglia, macrophages, and/or monocytes (Suppl. Table 3). Microglia markers in the top 20 genes include *Tyrobp*, which functions as an adaptor for a variety of immune receptors, *Cst3* (Cystatin C), *Anxa3* (Annexin A3), *Aif1* (Iba-1), and *Cd68*. OECs are known to endocytose bacteria and degrade axons, and therefore participate in the innate immune function (Khankan et al., 2016; Leung et al., 2008; Nazareth et al., 2015; Vincent et al., 2007). We also observed relatively high expression of genes involved in lysosome functions, such as *Cst3*, *Ctsz*, *Laptn5*, *Ctsb*, and *Lyz2*, compared to those in other OEC clusters (**Figure 4e**). Three additional top genes are involved in the complement system (*C1qa*, *C1qb*, and *C1qc*), and cluster 3 pathways are highly enriched for inflammatory response and neutrophil immunity (**Figure 4d**).

Smithson and Kawaja (2010) identified unique microglial/macrophages that immunolabeled with Iba-1 (*Aif1*) and Annexin A3 (*Anxa3*) in the olfactory nerve and outer nerve layer of the olfactory bulb. These authors proposed that Iba1-Anxa3 double-labeled cells were a distinct population of microglia/macrophages that protected the olfactory system against viral invasion into the cranial cavity. Based on our scRNA-seq data we offer an alternative interpretation that at least some of these Iba-1-Anxa3 cells may be a hybrid OEC-microglial cell type. Supporting this interpretation, there are a number of reports that suggest OECs frequently function as phagocytes (e.g., Khankan et al., 2016; Nazareth et al., 2020; Su et al. 2013).

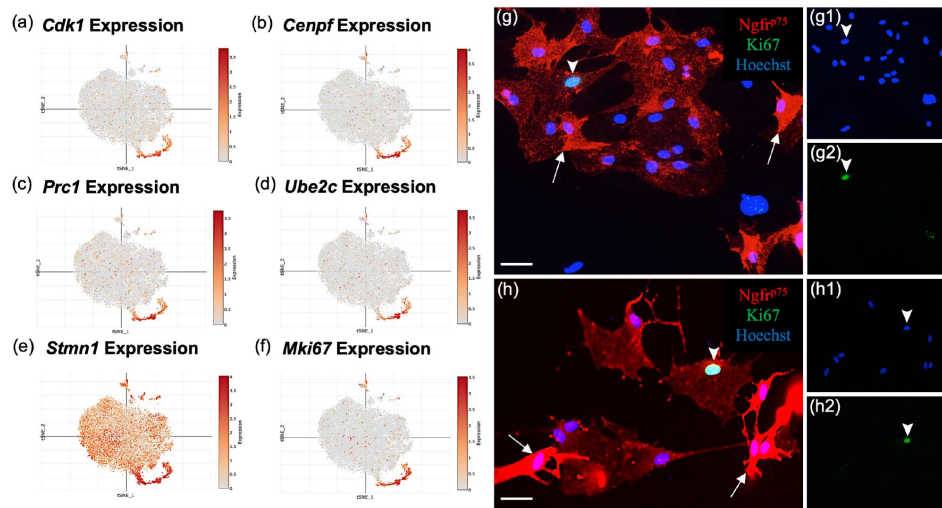


Figure 6.

Subcluster 2 is characterized by cell cycle and proliferative markers.

(a-f) These scRNA-seq plots show high expression of cell proliferation markers in cluster 2, supporting their function as OEC progenitor cells. Only *Stmn1* (e, Stathmin, a microtubule destabilizing protein) is broadly expressed across all five clusters. (g-g2, h-h2) Most cultured OECs are spindle-shaped and have high Ngfr^{p75} expression (red, white arrows). Of the OEC progenitors that express Ki67, 76% $\pm$ 8 of them display low levels of Ngfr^{p75} immunoreactivity and a “flat” morphology (g2, h2; green nuclei, arrowheads). The remainder of Ki67-expressing OECs express high levels of Ngfr^{p75} and are fusiform in shape (24% $\pm$ 8%, n=4 cultures, $p=0.023$). Hoechst marks all nuclei (g1, h1, blue nuclei, arrowheads). Scale bars: g, h = 50 μ m.

Cluster 4 has characteristics of astrocytes and oligodendrocytes

Cluster 4 is quite small. Its top two marker genes, *Sidt2*, a lysosomal membrane protein that digests macromolecules for reutilization and *Ckb*, a brain-type creatine kinase which fuels ATP-dependent cytoskeletal processes in CNS glia, are found in other OEC clusters. Four top genes, however, appear almost exclusively in cluster 4: 1) *Npvf* (Neuropeptide VF precursor/Gonadotropin-inhibitory hormone) inhibits gonadotropin secretion in several hypothalamic nuclei, and is expressed in the retina, 2) *Stmn2* (Stathmin2) is a tubulin-binding protein that regulates microtubule dynamics, 3) *Vgf* (non-acronymic) is synthesized by neurons and neuroendocrine cells and promotes oligodendrogenesis, and 4) *Mt3* (Metallothionein-3) is a small zinc binding protein associated with growth inhibition and copper and zinc homeostasis (Suppl. Table 3).

Other genes expressed by cluster 4 are either expressed by other glia and/or are ECM molecules. Five genes are associated with both oligodendrocytes and astrocytes (*Ckb*, *C1ql1*, *Gria2*, *Stmn2*, *Ntrk2*, Suppl. Table 3). Among these, the astrocytic gene *C1ql1* is a member of the Complement component 1q family and regulates synaptic connectivity by strengthening existing synapses and tagging inactive synapses for elimination. Astrocytic Thrombospondin 2 (*Thbs2*) also controls synaptogenesis and growth. Other genes reported in astrocytes include *Ptn* (Pleiotrophin), *Igfbp3* (Insulin-like growth factor binding protein 3), and *Mt3*, whereas *Gpm6b* (Glycoprotein M6b) is involved in the formation of nodes of Ranvier in oligodendrocyte and Schwann cell myelin. Cluster 4 also has a significant enrichment for genes involved in ECM organization (**Figure 4d** [↗](#); *Actn1*, *Col81a*, *Bgn*, *Fn1* and *Timp2*), together with clusters 0 and 1, as well as genes involved in positive regulation of cell migration together with cluster 3 (**Figure 4f** [↗](#)).

Trajectory and potential interactions among OEC subclusters

We performed pseudotime trajectory analysis using the Slingshot algorithm to infer lineage trajectories, cell plasticity and lineages by ordering cells in pseudotime based on their transcriptional progression reflected in our scRNA-seq data. Transcriptional progression refers to the changes in gene expression profiles of cells as they undergo differentiation or transition through different states. The trajectory analysis results suggest that there are potential transitions between specific OEC subclusters. Our results show that there are two distinct trajectories (**Figure 4g** [↗](#)). The first trajectory involves clusters 2, 0 and 4, whereas the second involves clusters 2, 0, 1, and 3. Although the directionality of these trajectories is indistinguishable, the fact that cluster 2 is enriched for cell cycle genes and is the converging point of both trajectories, suggests that cluster 2 proliferates into the other OEC clusters. The predicted trajectories based on our scRNAseq data suggest plasticity in the cell clusters.

We also modeled potential autocrine/paracrine ligand-receptor interactions between OEC subclusters using NicheNet (**Figure 4h** [↗](#)). This analysis revealed that clusters 0, 1, and 2 have more ligands going to other OEC clusters, whereas clusters 3 and 4 generally receive ligands from other clusters. The network also showed that clusters 0, 1 and 4 express genes encoding neurotropic factors such as *Bdnf*, *Cntf* and Neurturin (*Ntrn*), that are involved in neuronal survival and neurite outgrowth. *Gal* from cluster 1 is a mediator for glia-glia, and glia-neuron communication (Gresle et al., 2015 [↗](#); Ubink et al., 2003 [↗](#)). We found that cluster 4 secretes colony stimulating factor 1 (*Csf1*) which also shows potential regulation of cluster 3 through *Csf1r*. Overall, the trajectory and network analyses support potential cell plasticity and lineage relationships and cell-cell communications across OEC subclusters.

Spatial confirmation of defined OEC subclusters within the olfactory nerve layer *in situ*

Here we confirm, at the protein level, that some of the top genes derived from cultured OECs clusters are expressed in sections of the olfactory system from 8-10-week-old female Sprague Dawley rats. Our results illustrate the spatial distribution of 2-3 top genes from clusters 0 to 3 in the olfactory nerve layer (**Figure 7**; corresponding tSNE plots are in Suppl. Figure 2 for comparison). In cluster 0 we show high levels of Peripheral myelin protein 22 (Pmp22) in OECs (**Figure 7a**) with the remainder of the olfactory bulb unlabeled. The small protein Gliomedin (Gldn) is a component of Schwann cell microvilli and facilitates the formation of peripheral nodes of Ranvier (Eshed et al., 2005). Here we detect numerous small dot-like structures that overlay the olfactory nerve layer (**Figures 7b, c**). The detection of high levels of Gldn in nonmyelinating OECs without nodes of Ranvier is surprising and suggests that it may also function as a glial ligand associated with olfactory sensory neuron axons. For cluster 1, we examined the expression of the important axonal growth factor Activating transcription factor 3 (Atf3) and confirmed that OECs express it widely (**Figure 7d**). Growth associated protein 43 (Gap43), a well-established marker of OECs and immature olfactory sensory neurons, is highly expressed in the olfactory nerve layer and at a lower level in the glomeruli of the olfactory bulb (**Figure 7e**). Nestin (Nes), an intermediate filament associated with neural stem cells, is widely expressed in the olfactory nerve layer (**Figure 7f**).

Compared to the two large clusters, the remaining clusters contain more specialized OECs. The top marker in the proliferative cluster 2 is Stathmin1 (Stmn1), encoding a microtubule destabilizing protein found in areas of adult neurogenesis (Boekhoorn et al., 2014). Stathmin 1 immunoreactivity (**Figure 7g**, Suppl. Figure 2) is detected in all OEC clusters and immature olfactory sensory neurons, and therefore fills the olfactory nerve layer in a pattern similar to that seen in Gap43 (**Figure 7e**). Two markers associated with the cell cycle, Ube2c (**Figure 7h**) and Cdk1 (not shown), identify cluster 2 cells in the olfactory nerve layer, as well as the early progenitor marker Mki67 (**Figure 7i**). Cluster 3 is strongly associated with OEC immune function and the expression of Apolipoprotein E (ApoE), Annexin A3 (Anxa3), and Allograft inflammatory factor 1 (Aif1/Iba1) were confirmed. ApoE (**Figure 7j**) is detected throughout the olfactory nerve layer and includes some cellular structures. Expression of Anxa3 (**Figure 7k**) and Aif1/Iba1 (**Figure 7l**) is similar – distinct small cells labeled in the olfactory nerve layer. Our smallest group, cluster 4, has been difficult to confirm with selected antibodies targeting the top marker genes, likely due to the limited number of cells in this cluster. Overall, our *in-situ* experiments confirmed the presence and distribution of 4 out of the 5 OEC subtypes detected by scRNA-seq.

OECs synthesize and secrete Reelin

Reelin (*Reln*) is detected in OECs in this study (**Figures 2f**, 8a, g) and reported in previous microarray studies (Guerout et al., 2010; Roet et al., 2011). However, the presence of the *Reln* gene within OECs is disputed in the literature (Dairaghi et al., 2018; Schnauffer et al., 2009). *Reln* codes for a large ECM glycoprotein that is secreted by neurons (D'Arcangelo et al., 1995; Kubasak et al., 2004), but also by Schwann cells (Panteri et al., 2006). In the canonical Reelin-signaling pathway, Reelin binds to the very-low-density lipoprotein receptor (Vldlr) and apolipoprotein E receptor 2 (ApoER2) and induces Src-mediated tyrosine phosphorylation of the intracellular adaptor protein Disabled-1 (Dab1). Both Reelin and Dab1 are highly expressed in embryos and contribute to correct neuronal positioning. A recent study claimed that peripheral OECs express high levels of Dab1 and Vldlr (Dairaghi et al., 2018), and that Reelin was expressed only by mesenchymal cells located near the cribriform plate, not in OECs (Dairaghi et al., 2018; Schnauffer et al., 2009). Our scRNA-seq findings of OECs demonstrated low expression levels of *Dab1* and *Apoer2* (**Figure 8b-c**), moderate expression of *Vldlr* (**Figure 8d**) and substantial levels of *Reln* in OEC subclusters 0-3 (**Figure 8a**). Interestingly, OECs also contain proteolytic

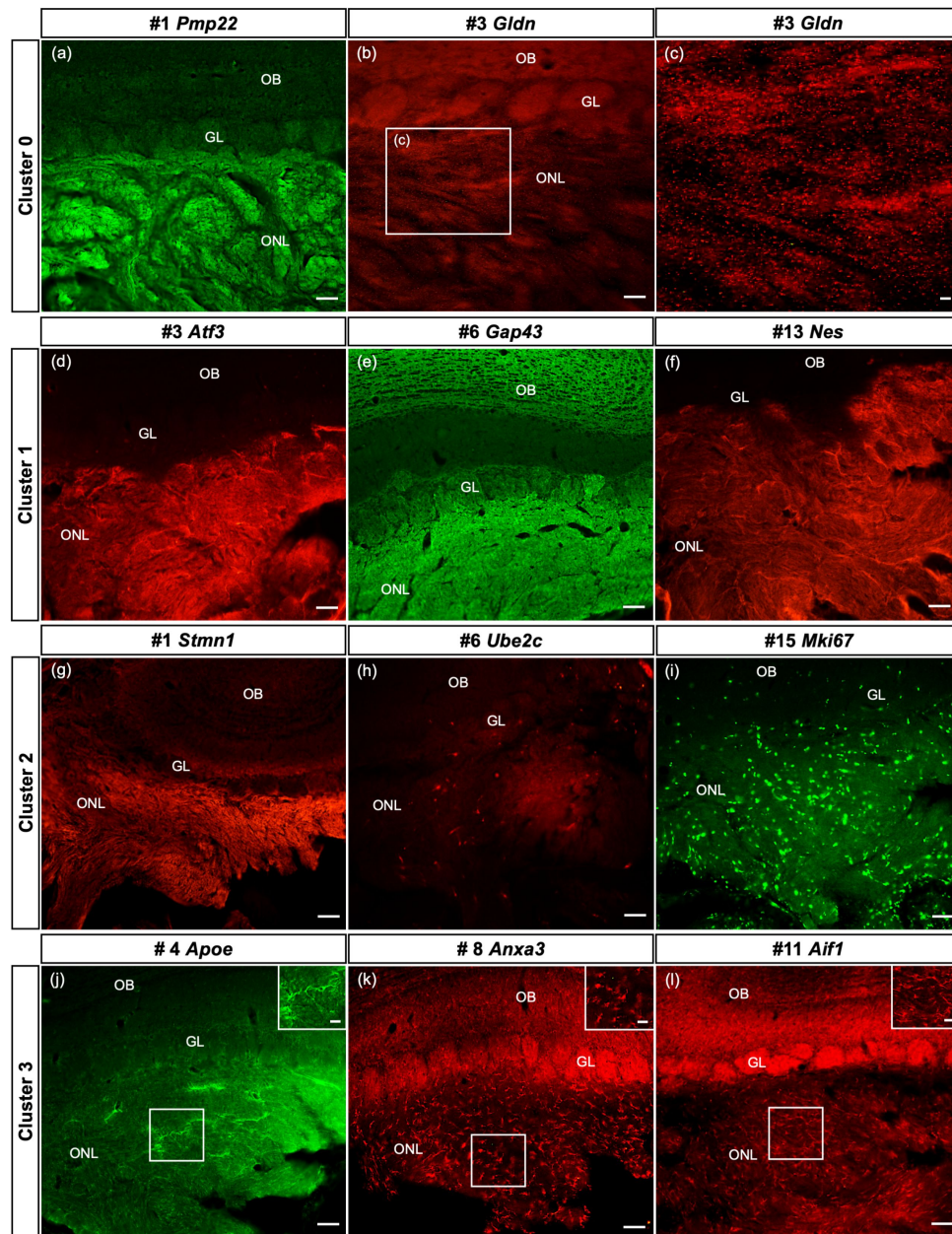


Figure 7.

Spatial confirmation of the defined OECs subclusters within the olfactory nerve layer.

The protein expression of a number of top 20 genes from this scRNA-seq study of purified OEC cultures is verified in olfactory bulb sections. In all images, the olfactory nerve layer (ONL, layer I) is at the bottom of the image, the glomeruli (GL, layer II) next, and the remainder of the olfactory bulb toward the top. Suppl. figure 2 illustrates the corresponding tSNE plots for each gene in [Figure 7](#). (a-c) The top gene in the largest cluster, Peripheral myelin protein 22, is highly expressed throughout the ONL. Gliomedin, the third-ranked gene in cluster 0, is detected as small discrete dot-like structures overlaying the ONL (b, box enlarged in c). (d-f) OECs in cluster 1 are immunolabeled by antibodies against the axonal growth factor *Atf3* (d) and the intermediate filament Nestin (f). High levels of *Gap43* in the ONL (e) are due to expression by OECs and axons of immature olfactory sensory neurons. (g-i) Strong immunoreactivity of Stathmin-1 in the ONL reflects labeled axons from immature olfactory sensory neurons and OECs. *Ube2c*, a G2/M cell cycle regulator, is expressed in a small number of cells in the ONL, whereas *Mki67*-labeled cells are widespread. (j-l) The immune function of this cluster is confirmed by antibodies against *Apoe* (j), *Anxa3* (k), and *Aif1/Iba1*, markers expressed by microglia and macrophages. Scale bars: a, b, d-l = 50 μ m; c, insets in j-l = 25 μ m.

enzymes that cleave the 400Kd secreted Reelin. Tissue plasminogen activator (tPA; **Figure 8e**) cleaves Reln at its C-terminus, and the matrix metalloproteinase ADAMTS-4 cuts at both the N- and C-cleavage sites of Reln (**Figure 8f**; Krstic et al., 2012).

The confusion about Reelin expression in OECs likely stemmed from the use of standard immunochemical methods to confirm its presence. Our attempts to detect Reelin expression in the adult olfactory nerve layer (ONL) or in cultured OECs were unsuccessful using several antibodies and multiple protocols, despite consistent localization of Reelin within mitral cells of the olfactory bulb. To understand this discrepancy, we examined Reelin expression in embryonic sections which included both the peripheral and OB-OECs. Images of E16.5 *Reln*^{+/+} olfactory system show Reelin expression in peripheral OECs and in neurons of the olfactory bulb (**Figure 8g**, green), but not on OB-OECs in the ONL or in sections of *Reln*^{-/-} mice (**Figure 8g**, g2, h). In comparison, *Blbp* expression is uniformly expressed in the ONL of both peripheral and OB-OECs (**Figure 8g** 1, h1, red). Enlargements of the region where axon bundles from olfactory sensory neurons join the ONL show that *Blbp* is present in OB-OECs throughout the ONL, but Reelin is detected only on peripheral OECs (**Figure 8g**, g1-2).

To determine if adult rat OB-OECs synthesize and secrete Reelin *in vitro*, we treated primary *Ngfr*^{p75}-purified OECs with or without Brefeldin-A, a fungal metabolite that specifically inhibits protein transport from the endoplasmic reticulum to the Golgi apparatus (Misumi et al., 1986). In Brefeldin-treated primary OEC cultures, spindle-shaped GFP-labeled OECs contain Reelin in their cytoplasm (**Figure 9a**, arrows). Large contaminating Reln-immunonegative cells present in primary cultures serve as an intrinsic control (**Figure 9a**, arrowhead). Reelin expression was also confirmed in Brefeldin-treated *Ngfr*^{p75}-purified OEC cultures (**Figure 9b-c**), a finding which supports our scRNA-seq results.

To further confirm that cultured OECs express and secrete Reelin, we performed western blotting. The G10 antibody recognizes full-length Reelin protein, which is approximately 400 kDa, and its two cleaved fragments at ~300 and ~180 kDa (Lambert de Rouvroit et al., 1999). Both the rat ONL and *Reln*^{+/+} mouse olfactory bulbs showed the characteristic large molecular weight band at 400 kDa and another at 150 kDa (**Figure 9d**, lanes 1-2, Reln). As expected, these bands were absent in mutant *Reln*^{-/-} olfactory bulbs (**Figure 9d**, lane 3). Additionally, Reelin-positive bands were present in blots of OEC whole cell lysates (WCL), representing Reelin synthesized in OECs, and OEC conditioned media (CM; **Figure 9d**, lanes 4-8 Reln), representing Reelin secreted by OECs. A 4-15% gradient gel allowed for the visualization of all three Reelin isoforms (**Figure 9e**). Brain extracts of *Reln*^{+/+} (positive control) and *Reln*^{-/-} (negative control) mice confirm the specificity of Reelin detection (**Figure 9e**, lanes 1-2). All three Reelin isoforms at their corresponding molecular weights are present in rat ONL and OEC CM samples (**Figure 9e**, lanes 3-6). These results clearly indicate that cultured OECs derived from the ONL can synthesize and secrete Reelin.

Discussion

Our scRNA-seq study of immunopurified OECs identified five OEC subtypes with distinct gene expression patterns, pathways, and properties of each subtype. The well-established OEC markers are widely distributed throughout most clusters derived from cultured OECs. We also experimentally confirmed a number of the top markers at the protein level in the OEC subtypes on sections of the rat olfactory system and established that OECs secrete both Reelin and Ctgf, ECM molecules reported to be important for neural repair and axon outgrowth. Additional proregenerative OEC marker genes such as *Atf3*, *Btc*, and *Pcsk1* also were identified. Our findings provide an unbiased and in-depth view of the heterogeneity of OEC populations with demonstrated injury repair capacity, and the potential mechanisms that underlie their regenerative and protective properties.

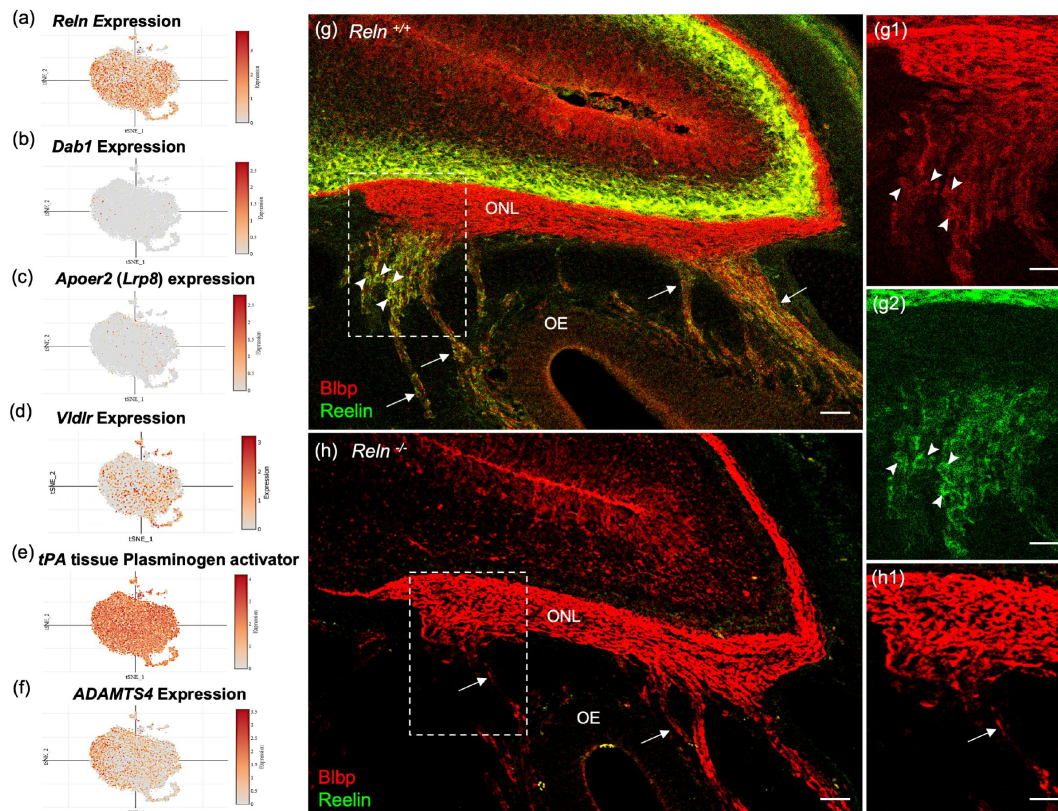


Figure 8.

Reelin-signaling pathway marker genes in OEC clusters and Reelin immunoreactivity in embryonic olfactory system.

(a-f) These six tSNE plots show OEC gene expression levels associated with Reelin signaling. *Reln* is highly expressed in most OEC clusters (a), *Disabled-1 (Dab1)* (b), and *Apolipoprotein E receptor 2 (Apoer2)* (c) show little expression, and *Very-low-density lipoprotein receptor* has moderate expression (*Vldlr*, d). The serine protease tissue Plasminogen Activator (e, *tPA*) is highly expressed in all OEC subsets and cleaves secreted Reelin only at its C-terminus. ADAMTS-4 cleaves both the N- and C-terminals of Reelin, and is expressed at low levels by OECs (f). **(g-g2)** Sagittal section of the olfactory epithelium (OE) and olfactory bulb from an E16.5 *Reln*^{+/+} mouse immunolabeled for Blbp (g, g1; red) and Reelin (g, g2; green). Axons of olfactory sensory neurons (arrows) are surrounded by peripheral OECs that express both Blbp (g1, arrowheads) and Reelin (g2, arrowheads). High *Reln* expression is obvious in olfactory bulb neurons (g, g1, green) but the olfactory nerve layer (ONL) only expresses Blbp (g, g2). **(h, h1)** No Reelin is detected in a sagittal section from a *Reln*^{-/-} mouse, yet Blbp expression in the ONL appears normal. Scale bars: g-h = 50 μ m; g1, g2 = 50 μ m.

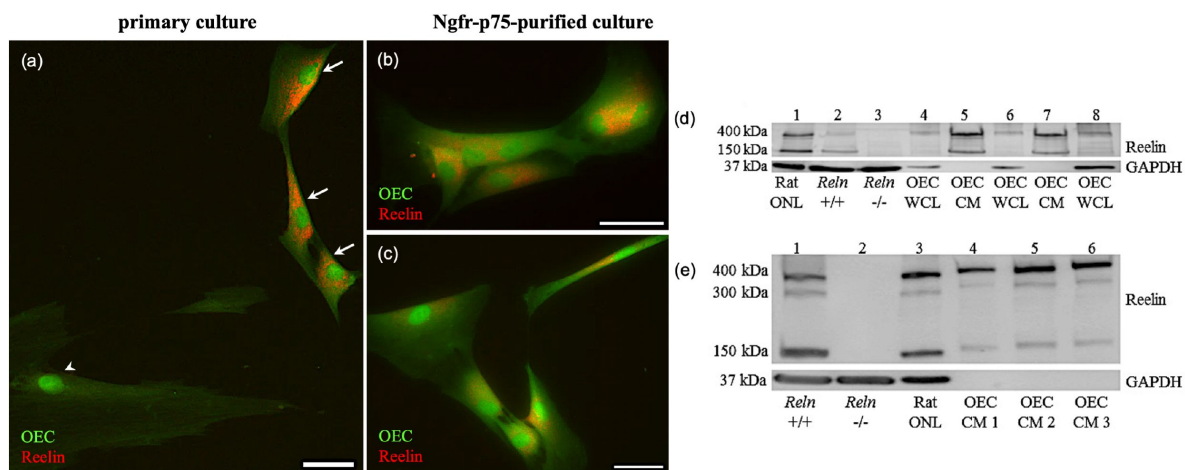


Figure 9.

Reelin is expressed and secreted by olfactory ensheathing cells.

(a-c) Rat OEC cultures were treated with Brefeldin-A to inhibit protein transport and subsequent secretion. Reelin immunoreactivity (red) was detected in GFP-labeled OECs (a, arrows), but not in another cell type (arrowhead) in this primary culture. GFP-labeled OECs which were immunopurified with anti-Ngfr^{p75} also express Reelin (red; b, c). (d) Western blot confirms the expression of Reelin in rat olfactory nerve layer I and II (ONL; lane 1 of western blot). *Reln*^{+/+} and *Reln*^{-/-} mouse olfactory bulbs were used as positive and negative controls, respectively (lanes: 2 and 3). Reelin synthesized by cultured OECs was found in whole cell lysates (WCL; lanes: 4, 6, and 8), whereas Reelin secreted by cultured OECs into tissue culture medium was measured in OEC "conditioned medium" (CM; lanes: 5 and 7). GAPDH was the loading control for tissue homogenates (lanes 1-4, 6, 8). (e) All three Reelin isoforms (400, 300, and 150 kDa) were visualized using a 4-15% gradient gel. *Reln*^{+/+} and *Reln*^{-/-} mouse cortices were used as controls (lanes: 1 and 2). Reelin was detected in the rat ONL (lane 3) and in three rat OEC-CM samples (lanes: 4, 5, and 6). GAPDH was the loading control for tissue homogenates (lanes: 1-3). Scale bars: a-c = 40 μ m.

This scRNA-seq analysis confirmed the immunopurification of the OECs compared with the mixture of OECs grown with numerous fibroblasts and microglia. Although some may debate if purified OECs are needed to maximize neural repair, this study suggests that immunopurified OECs have higher expression of anti-stress genes than unpurified controls. Several of our earlier studies, which transplanted similarly purified OECs, also showed evidence of neural regeneration following severe spinal cord injury (Khankan et al., 2016 [DOI](#); Takeoka et al., 2011 [DOI](#); Thornton et al., 2018 [DOI](#)). In addition to the purity of OECs, we found significant overlap of the marker genes between our study and those from the meta-analysis of five OEC microarray studies on cultured OB-OECs (Roet et al., 2011 [DOI](#)), supporting the reproducibility of our scRNA-seq findings. When we compared our data from cultured OB-OECs to previous scRNA-seq studies of the olfactory system tissue which included peripheral OECs (Durante et al, 2000), or a mixture of peripheral and OB-OECs (Tepe et al., 2018 [DOI](#)), there was little overlap. Most likely, the substantial differences were due to the dissection areas, techniques used for OEC preparation, the species studied, as well as sex and age of the subjects. It is important to note that our OB-OEC populations were repeatedly shown to induce injury repair, and their subtypes and markers support their diverse reparative functions, whereas the neural repair function of the cell populations in the other scRNA-seq studies using different tissue preparations is unclear.

OECs are hybrid glia that promote neural repair

Based on intrinsic gene expression patterns revealed by scRNA-seq data, there is expression of select markers for Schwann cells, astrocytes, oligodendrocytes and microglia within our five clusters of OECs. We observed transcriptionally distinct clusters that highly expressed select markers for proliferation (cluster 2), microglia (cluster 3), and astrocytes and oligodendrocytes (cluster 4). The microglia/macrophage-like cells reported by [Smithson and Kawaja \(2010\) DOI](#) do resemble our transcriptome-based subtyping of OEC cluster 3. Our results further support the activities of OECs in modulating immune responses, based on the expression of genes involved in the endocytosis of bacteria, the regulation of microglia activation, and the expression of numerous lysosomal pathways ([Leung et al., 2008 DOI](#)). OECs in cluster 4 are distinctive by their expression of several astrocytic markers that regulate synaptogenesis (Thrombospondin 2) and synaptic connectivity, and remove inactive synaptic connections (Complement component 1q family; [Christopherson et al., 2005 DOI](#); [Eroglu & Barres, 2010 DOI](#)). Additionally, our trajectory and network analyses indicate potential relationships between the OEC subclusters, which warrant further experimental testing.

Both OECs and Schwann cells originate from the neural crest, express similar immunological markers, and share many transcriptional similarities ([Forni et al., 2011 DOI](#); [Vincent et al., 2005 DOI](#)). OECs share phenotypic characteristics with myelin-producing Schwann cells and oligodendrocytes, but more closely resemble Schwann cells ([Doucette, 1991 DOI](#); [Radtko et al., 2011 DOI](#); [Ramon-Cueto & Valverde, 1995 DOI](#)). Gliomedin is a required ECM protein that initiates the assembly of nodes of Ranvier along Schwann cells ([Eshed et al., 2005 DOI](#)), yet despite the fact that OECs do not have nodes of Ranvier, they express high levels of Gliomedin. This novel Gliomedin expression in OECs suggests a possible function as a glial ligand for the unmyelinated axon bundles that OECs ensheath. Following peripheral nerve injury, the loss of contact between axons and myelinating Schwann cells induces a nonmyelinating phenotypic change in Schwann cells ([Fu & Gordon, 1997 DOI](#)). These dedifferentiated nonmyelinating Schwann cells are proregenerative and upregulate the expression of Glial fibrillary acidic protein (Gfap; ([Jessen et al., 1990 DOI](#)), [Ngfr^{p75}](#) receptor ([You et al., 1997 DOI](#)), cell adhesion molecules L1 and N-Cam ([Martini & Schachner, 1988 DOI](#)), and neurotrophic factors, including Bdnf ([Meyer et al., 1992 DOI](#)) and Glial cell-line derived neurotrophic factor ([Trupp et al., 1995 DOI](#)). Genes for these proregenerative factors are detected in OECs in the present study, as well as the transcription factor *Mitf*, that controls a network of genes related to plasticity and repair in Schwann cells ([Daboussi et al., 2023 DOI](#)).

Our *in vitro* and *in vivo* experimental validation of numerous marker genes at the protein level strongly support their validity. Based on our tissue localization, the two large clusters (0 and 1) were found throughout the olfactory nerve layer, whereas cluster 2 (progenitors) and cluster 3 (microglial) are scattered within the olfactory nerve layer. Future functional studies of these subclusters will benefit from both the marker genes and their localizations uncovered in this study.

Reelin and CTGF are distinctive markers for OECs

Reelin is a developmentally expressed protein detected in specific neurons, in addition to OECs and Schwann cells. The canonical Reelin-signaling pathway involves neuronal-secreted Reelin binding to Vldlr and ApoER2 expressed on Dab1-labeled neurons. Following Reelin binding, Dab1 is phosphorylated by Src family kinases which initiates multiple downstream pathways. Very little is known, however, about Reelin secreted by glia. Panteri et al. (2006) [reported](#) that Schwann cells express low levels of Reelin in adults, and that it is upregulated following a peripheral nerve crush, as is reported above for many neurotrophic factors. Reelin loss in Schwann cells reduced the diameter of small myelinated axons but did not affect unmyelinated axons (Panteri et al., 2005). In the olfactory system, OECs ensheath the Dab1-labeled, unmyelinated axons of olfactory sensory neurons which are continuously generated and die throughout life. OEC transplantation following spinal cord injury would provide an exogenous source of Reelin that could phosphorylate Dab1-containing neurons or their axons. Dab1 is expressed at high levels in the axons of some projection neurons, such as the corticospinal pathway (Abadesco et al., 2014 [reported](#)). Future experiments are needed to explore the function that glial-secreted Reelin may have on axonal regeneration.

Adult mammals show little evidence of spontaneous axonal regeneration after a severe spinal cord injury in contrast to transected neonatal rats (Bregman, 1987 [reported](#); Bregman et al., 1993 [reported](#)) and young postnatal opossums (Lane et al., 2007 [reported](#)). In immature mammals, axons continue to project across or bridge the spinal cord transection site during development. Lower organisms such as fish, show even more evidence of regeneration following severe SCI. Mokalled et al. (2016) [reported](#) that glial secretion of Ctgfa/Ccn2 was both necessary and sufficient to stimulate a glial bridge for axon regeneration across the zebrafish transection site. Cells in the injury site that express Ctgf include ependymal cells, endothelial cells, and reactive astrocytes (Conrad et al., 2005 [reported](#); Mokalled et al., 2016 [reported](#); Schwab et al., 2001 [reported](#)). Here we show that, although rare, Ctgf-positive OECs can contribute to glial bridge formation in adult rats. The most consistent finding among our severe SCI studies combined with OEC transplantation is the extent of remodeling of the injury site and axons growing into the inhibitory lesion site, together with OECs and astrocytes. The formation of a glial bridge across the injury was critical to the spontaneous axon generation seen in zebrafish (Mokalled et al., 2016 [reported](#)) and likely contributed to the axon regeneration detected in our OEC transplanted, transected rats (Dixie, 2019 [reported](#); Khankan et al., 2016 [reported](#); Takeoka et al., 2011 [reported](#); Thornton et al., 2018 [reported](#)).

Limitations in these OEC scRNA-seq studies

We recognize that this study is a single snapshot of OEC gene expression derived from adult female rats before they are transplanted above and below the spinal cord transection site. We would expect the gene expression of transplanted OECs to change in each new environment, i.e. as they migrate into the injury site, integrate into the glial scar, and wrap around axons. Based on our past studies, OECs survived in an outbred Sprague-Dawley female rat model for ~ 4 weeks (Khankan et al., 2016 [reported](#)) and in an inbred Fischer 344 female model for 5-6 months (Dixie, 2019 [reported](#)). As spinal cord injury transplant procedures are further enhanced and OEC survival improves, these hybrid glial cells should be examined at multiple time points after transplantation to better evaluate their proregenerative characteristics.

Due to the extensive urinary tract dysfunction in spinal cord transected rats, most studies are conducted on females as their short urethra facilitates daily manual bladder expression. Our study, therefore, was carried out only on adult female rats, so sex differences and the generalizability of our findings to adult male rats would require further investigation. We also did not modulate any of the genes or proteins in the identified OEC subtypes to test their causal and functional roles, thus our findings remain correlative. Future gene/protein modulation studies are necessary to understand the functional roles of the individual OEC subtypes in the context of their reparative functions to determine which pathways and subtypes are more critical and can be enhanced for neural repair. Our current findings build the foundation for these future studies to help resolve the role of OECs in spinal cord injury repair.

Extensive differences between OEC preparations contribute to the large variation in results from OEC treatments following spinal cord injury. This scRNA-seq study focused entirely on OB-OECs, and the next step would be to carry out similar studies on the peripheral, lamina-propria-derived OECs to discern the differences between the two OEC populations. Such comparative studies using scRNA-seq will help define the underlying mechanisms and resolve the variability in results from OEC-based therapy. Detailed studies of the composition of different OEC transplant types will contribute to identifying the most reparative cell transplantation treatments.

Conclusion

To understand how immunopurified OECs repair spinal cord injuries, we carried out an unbiased scRNA-seq analysis and found that OECs have a wide range of subtypes including a proliferative state, microglia-like cells and neural regenerative clusters. We demonstrated that these multi-functional OECs are clearly distinguishable from each other by expression of unique genes and pathways within subtypes based on the scRNAseq data and our *in vivo* tissue studies. These subtypes of OECs produce important ECM genes, including *Reln* and *Ctgf* and many other signaling molecules which are involved in a wide range of intercellular communication as well as axon regeneration. Our single-cell resolution investigation offers a comprehensive molecular landscape of OECs and numerous novel targets for future injury repair and functional/mechanistic studies.

Material and methods

Animals

All of the experimental procedures used on animals were approved by the Chancellor's Animal Research Committee at UCLA and carried out in accordance with the National Institutes of Health guidelines. Animals were housed in the Terasaki Life Science Building vivarium under standard conditions with free access to food and water.

Sprague-Dawley rats

We bred our male GFP-expressing Sprague-Dawley rats (Perry et al, 1999) with unlabeled female Sprague-Dawley rats (Charles Rivers Laboratory), which were used previously to obtain OECs from female rats for two of our spinal cord injury studies (Khankan et al., 2016 [DOI](#); Thornton et al., 2018 [DOI](#)) in order to unambiguously identify the transplanted GFP-positive OECs near the injury sites. A total of 8, 8-10-week-old GFP-labeled female rats were used to obtain GFP-OECs for scRNA-seq. Only females were used in order to match the sex of previous SCI studies conducted exclusively on female rats (Dixie, 2019 [DOI](#); Khankan et al., 2016 [DOI](#); Takeoka et al., 2011 [DOI](#); Thornton et al., 2018 [DOI](#)). Following complete thoracic spinal cord transection, an adult rat is unable to urinate voluntarily and therefore urine must be manually “expressed” twice a day throughout the experiment. Females have a shorter urethra than males, and thus their bladders are easier to empty completely. An additional 4, 8-10-week GFP-labeled rats were used to isolate Reelin protein for western blotting and 6, 8-10-week unlabeled female Sprague-Dawley rats were used to confirm

gene expression in the five OEC subtypes. An overdose of ketamine-xylazine or pentobarbital was used for euthanasia before either the rat olfactory bulbs were removed or the rats were perfused with 4% paraformaldehyde or 2% PLP (2% Paraformaldehyde; 0.075M Lysine-HCL-monohydrate; 0.010M Sodium periodate; 0.1M sodium phosphate) followed by a 1-2 hr postfix.

Reeler mice

To study Reelin expression in OECs, heterozygous (*Reln*^{+/-}) mice, originally purchased from the Jackson Laboratory (B6C3Fe-*ala-Reln*^{fl}, Bar Harbor, ME), were used for breeding. Genotypes of these mice were identified by PCR (D'Arcangelo et al., 1996 [DOI](#)). To obtain timed pregnancies, *Reln*^{+/-} mice were bred and checked every morning for plugs. Pregnant mice were heavily anesthetized with pentobarbital (100 mg/kg) and embryos were extracted at embryonic day 16.5. Heads were immersed in 2% PLP, washed in PB, cryoprotected, and embedded in Shandon M1 embedding matrix (Thermo Scientific, 1310).

Preparation of olfactory bulb-derived OEC cultures from rats and mice

Olfactory bulbs were collected from 8-10-week-old GFP-labeled Sprague-Dawley rats or *Reln*^{+/+} and *Reln*^{-/-} mice. The leptomeninges were removed to reduce fibroblast contamination. Methods to prepare OEC primary cultures were adapted from Ramon-Cueto et al. (2000) [DOI](#) and follow those reported by Runyan and Phelps (2009) [DOI](#) for mouse OECs and Khankan et al. (2015) for rat OECs. For scRNA-seq, this protocol was replicated on two separate dates and used a total of 8 adult female Sprague-Dawley rats. Primary cultures were generated from both olfactory bulbs from each rodent. The first two layers of the olfactory bulb were dissected, isolated, and then washed in Hank's balanced salt solution (HBSS, Gibco, Rockville, MD) prior to tissue centrifugation at 365 g for 5 mins. The tissue pellet was resuspended in 0.1% trypsin and HBSS without Ca²⁺/Mg²⁺ (Gibco, Rockville, MD), then placed in a 37°C water bath, and mixed intermittently for 10 mins. A mixture of 1:1 Dulbecco's modified eagle medium (DMEM) and Ham's F12 (D/F medium, Gibco) supplemented with 10-15% Fetal Bovine Serum (FBS, Hyclone, Logan, UT) and 1% Penicillin/Streptomycin (P/S, Gibco; D/F-FBS-P/S medium) was used to inactivate trypsin prior to centrifugation. Dissociated cells were rinsed, centrifuged 3 times, and plated into 12.5 cm² culture flasks pre-coated with 0.05mg/ml poly-L-lysine (PLL, Sigma, St. Louis, MO). Cells were maintained at 37°C for 5-8 days and D/F-FBS-P/S medium was changed every 2 days.

Immunopurification was carried out using hydrophobic petri dishes coated overnight with Biotin-SP-conjugated AffiniPure goat anti-rabbit IgG (1:1000; Jackson ImmunoResearch, West Grove, PA) in 50 mM Tris buffer at 4°C followed by another overnight incubation at 4°C with either mouse anti-Ngfr^{p75} for rat OECs (Suppl. Table 4, Chandler et al., 1984 [DOI](#)) or rabbit anti-Ngfr^{p75} for mouse OECs (Suppl. Table 4) in 25mM Phosphate-buffered saline (PBS). Dishes were rinsed 3 times with 25 mM PBS and treated with a mixture of PBS and 0.5% Bovine serum albumin (BSA) for 1 hr at room temperature. Prior to the addition of cells, antibody-treated dishes were washed with PBS and DMEM.

In preparation for immunopanning, two pairs of adult rat olfactory bulbs (for scRNA-seq) or four pairs of mouse bulbs (for western blots) were dissociated with 0.25% trypsin-EDTA at 37°C for 3 mins and D/F-FBS-P/S medium was added to inactivate trypsin. Following a medium rinse and centrifugation, resuspended cells were seeded onto pre-treated mouse or rabbit anti-Ngfr^{p75} petri dishes and incubated at 37°C for 10 mins. Unbound rat cells were removed with medium and saved as 'leftover' control cells for scRNA-seq. A cell scraper was used to recover bound cells that were then subjected to a second round of immunopanning in which purified cells from 2 rats on each experimental day were combined yielding 2 samples of purified Ngfr^{p75}-positive rat OECs and 2 samples of 'leftover' control cells. All cells for sequencing were replated into PLL-coated flasks. Immunopurified Ngfr^{p75}-positive mouse OECs were plated on PLL-coated 4-chamber

polystyrene-vessel culture slides (BD Falcon, San Jose, CA) or used for western blots. Purified OECs and 'leftover' controls were incubated at 37°C with 5% CO₂ for 7-8 days in D/F-FBS-P/S medium supplemented with a mitogen mixture of pituitary extract (20 µg/ml, Gibco) and forskolin (2 µM, Sigma). Media was changed every 2 days and mitogens were withdrawn 2 days prior to harvesting these cells for scRNA-seq.

Brefeldin-A treatment

Rat OECs that contained GFP could be visualized directly. Primary and purified GFP-expressing OEC cultures were rinsed 3 times with D/F medium prior to treatment with Brefeldin-A to block protein transport from the endoplasmic reticulum to the Golgi apparatus and thus block secretion. Brefeldin-A (5 µg/ml; Epicenter, Madison, WI) was mixed with D/F medium and added to OEC cultures for 2 hrs at 37°C while the control cultures received D/F medium only. Cultures were fixed with cold 4% paraformaldehyde in 0.1 M PB for 15 mins at RT, rinsed 3 times with PBS, and stored in PBS with sodium azide at 4°C until immunostaining.

Detection of Reelin expression in rat and mouse OEC cultures, tissue sections and western blots

Cultured rat OECs treated with or without Brefeldin-A were labeled with mouse anti-Reelin G10 and rabbit anti-*Ngfr*^{p75} (Suppl. Table 4). Culture slides were rinsed with phosphate-buffered saline (PBS; 0.1M phosphate buffer; 0.9% NaCl), blocked with 5% normal goat serum for 1 h, and incubated with the appropriate primary antibodies overnight. The following day, slides were rinsed 3 times with PBS, incubated with species-appropriate Alexa Fluor 594 and/or 647 (1:500, 1:100; Jackson ImmunoResearch) for 1 hr at RT, and then cover slipped with Fluorogel (Electron Microscopy Sciences, Hatfield, PA).

Sagittal sections of the mouse E16.5 olfactory system were cut 40 µm thick, slide mounted, and stored in PB with 0.06% sodium azide. For Reelin and *Blbp* detection, sections were incubated overnight at RT in goat-anti-Reelin (Suppl. Table 4), rinsed thoroughly, and incubated with donkey anti-goat 488 (1:100; Jackson ImmunoResearch, #705-545-0030).

Sections were then incubated in rabbit polyclonal anti-*Blbp* (Suppl. Table 4) overnight followed by 1 hr in donkey anti-rabbit 555 (1:800; Life Technologies, #A31572) and coverslipped with Fluorogel. Confocal images were obtained from a Zeiss Laser Scanning Microscope (LSM800) with solid-state lasers 488 and 561 nm for double-labeled images. Spinal cord sections from [Khankan et al., 2016](#) were sectioned sagittally at 25 µm thick.

Tissue harvesting, protein isolation, and western blotting

Olfactory bulbs from adult *Reln*^{+/+} and *Reln*^{-/-} mice and olfactory nerve layers from adult Sprague Dawley rats were freshly dissected as described for primary OEC cultures. Olfactory tissue was collected, placed on ice, and homogenized in RIPA lysis buffer plus protease inhibitor cocktail (Sigma). Protein concentrations of brain and olfactory extracts were determined using the RC DC Protein Assay kit (Bio-Rad, Hercules, CA) as described ([Miller et al., 2008](#)). Whole-cell lysates and conditioned medium (CM) were obtained from purified OEC cultures maintained in D/F medium. CM was collected from OEC cultures before cells were incubated with RIPA lysis buffer and collected with a cell scraper. Protein samples were stored at -80°C.

Protein samples (50 µg) were heated to 90–100°C for 3 mins, resolved on 10% SDS-PAGE or 4–15% mini-protean TGX gels (Bio-Rad) run in Tris-Glycine-SDS (TGS) buffer (Bio-Rad) at 60V, and then transferred onto 0.22 µm PVDF/nitrocellulose membranes (Bio-Rad) in TGS with 20% methanol at 25V overnight at 4°C. Precision plus protein standards in dual color (Bio-Rad) were used to determine molecular weight. Membranes were cut, blocked with 5% non-fat dry skimmed milk (Bio-Rad) in Tris-buffered saline containing 0.1% Tween-20 (TBST) for 1 hr at room temperature on

a shaker, and then incubated overnight at 4°C TBST plus 2.5% milk containing anti-Reelin G10 or loading control mouse anti-GAPDH antibodies (Suppl. Table 4). Following primary antibody incubation, blots were washed 3 times in TBST for 10 mins while shaking and then probed with horseradish peroxidase-conjugated secondary antibodies anti-mouse IgG (1:2,000 for Reelin, 1:5,000 for GAPDH) in TBST. Immunoblots were developed using a chemiluminescence HRP detection kit (GE Healthcare, Piscataway, NJ,) and imaged with a Typhoon scanner (GE Healthcare).

Single-cell RNA-sequencing

Cell collection and culture

Flasks with purified OECs and leftover cells were rinsed with HBSS and dissociated with trypsin-EDTA as above. Cells were washed in D/F-FBS-P/S medium and further resuspended with siliconized pipettes. Media and cells were washed and centrifuged at 365 *g* for 10 min, and resuspended. After the final wash, cells for sequencing were resuspended in PBS+BSA in volumes of 1200 cells/μL. On each of 2 different culture days, 4 flasks of cells (2 purified OECs and 2 ‘leftovers’) were generated for scRNA-seq, yielding 8 samples in total (4 purified OECs and 4 ‘leftovers’). To confirm our scRNA-seq results directly with immunocytochemistry, we plated extra cells from the second preparation on PLL-coated coverslips in 24-well plates with 500μL D/F-FBS-P/S medium. Medium was changed at 2 days, cultures were fixed at 3 days in cold 4% paraformaldehyde for 20 mins, washed 3X with PB and stored in PB with 0.06% sodium azide.

scRNA-seq was performed using the 10X Genomics Chromium scRNA-seq system which incorporates microfluidic capture-based encapsulation, barcoding, and library preparation (10X Genomics, Pleasanton, CA). About 10,000 cells per OEC or leftover preparation were loaded into a Chromium Chip B along with partitioning oil, reverse transcription reagents, and a mix of hydrogel beads containing 3,500,000 unique 10X barcodes. Paired-end sequencing was performed on a Novaseq S4 system, using the v3 Illumina platform, at 20-50K reads/cell. The paired-end reads were processed using the Cell Ranger pipeline and *Rattus norvegicus* (Rnor_6.0) from Ensembl as the reference genome (Yates et al., 2020 [DOI](#)) to produce a gene expression matrix. Among the 8 samples, one OEC sample showed a low fraction of reads and RNA content in cells, which may indicate high level of ambient RNA (Suppl. Figure 3a). This sample was removed from further analysis, leaving 3 OEC samples and 4 leftover samples for downstream analysis.

scRNA-seq data quality control, normalization, and integration

Gene expression matrix from each of the 7 remaining samples that passed quality control was loaded into the Seurat R package ver 3.0 (Stuart et al., 2019 [DOI](#)). Cells were called based on the number of genes (200-5000), unique molecular identifiers (UMIs; 5000-20000), and percent of mitochondrial genes (<20%; Suppl. Figure 3b). After quality control, log normalization was performed within each sample using *NormalizeData* function with default parameters. Top 2,000 variable genes were selected using *FindVariableFeatures*. The seven samples were integrated together with *FindIntegrationAnchors* and *IntegrateData* functions which incorporate canonical correlation analysis to align cells with similar transcriptomic patterns across samples.

Cell clustering and cell type identification

The integrated Seurat object was used for principal component analysis (PCA). The top 15 PCs were used to construct the k-nearest neighbor graph, followed by Louvain algorithm to cluster cells based on similar gene expression patterns. Cell clusters were visualized using tSNE plots.

To assign cell type identities to each cluster, marker genes of each cluster were found using *FindAllMarkers* with average log fold change > 0.5 and minimum percent difference > 0.25. Cell cluster marker genes were subsequently compared to known OEC cell type markers curated from

previous studies that used different species, sex, age, dissection areas, and cell preparation methods (Durante et al., 2020 [↗](#); Tepe et al., 2018 [↗](#)). Additional marker genes for fibroblasts and multiple glial cell types including astrocytes, oligodendrocytes, and microglia were also used to compare with those of the cell clusters. Cell clusters whose marker genes match known cell type markers were labeled with the corresponding cell type identity. Gene expression heatmaps of top markers were generated to visualize the distinct expression patterns of the marker genes for different cell clusters.

OEC subtype identification

A subset of cells expressing well-known OEC markers (e.g., *Ngfr*^{p75}, *S100β*, and *Sox10*) was labeled as OECs and then selected for further subclustering analysis to identify potential OEC subtypes. To do this, OECs from the original Seurat object were extracted using the *subset* function and the top 10 PCs of each OEC subset were used for clustering analysis and tSNE visualization. Marker genes for each OEC subcluster were identified using the *FindAllMarkerGene* function. Pseudotime trajectory analysis across the OEC subtypes was performed using Slingshot version 1.8.0 (Street et al., 2018 [↗](#)).

Pathway enrichment analysis

To annotate the biological functions of the marker genes of individual cell clusters or OEC subclusters, we carried out pathway enrichment analysis using EnrichR ver 3.0, where KEGG, Reactome, and GO Biological Process were used as reference pathway databases (Chen et al., 2013 [↗](#)). Top pathways were ranked by multiple-testing adjusted P-values.

Potential cell-cell ligand-receptor interactions across OEC subtypes

In order to characterize potential interactions mediated by secreted ligand-receptor pairs between subtypes of OECs, we implemented NicheNet (Browaeys et al., 2020 [↗](#)), which curates ligand-receptor interactions from various publicly available resources. The NicheNet ligand-receptor model was integrated with markers from each OEC subtype to find potentially activated ligand-receptor pairs. Differentially expressed ligands identified by markers from source OEC subtypes and activated receptors identified by NicheNet in target OEC subtypes were used to assess ligand-receptor interaction. Ligand-receptor interaction networks between cell types were visualized using Cytoscape (Shannon et al., 2003 [↗](#)).

Confirmation of OEC gene expression *in vitro* and *in vivo*

Immunocytochemistry experiments to confirm gene expression revealed by scRNA-seq were conducted on extra cultured OECs and ‘leftover’ controls which were replated and cultured for 3 additional days. OECs samples were treated with the following primary antibodies against OEC marker genes identified from scRNA-seq with experimental parameters recorded in Suppl. Table 4: 1) polyclonal and monoclonal anti-Blbp, 2) anti-N-Cadherins, 3) anti-green fluorescent protein, 4) anti-Ki67, 5) anti-L1, 6) anti-NCAM, 7) monoclonal anti-*Ngfr*^{p75}, 8) anti-S100β, and 9) anti-SOX10. Prior to adding the primary antibodies, the cells were incubated in 0.3% Triton-X 100 detergent in PBS. Cells were blocked in 5% normal donkey serum (Jackson ImmunoResearch Laboratories, # 017-000-121) with 0.3% Triton-X before adding the primary antibodies and left to incubate at RT overnight on the rotator. Following multiple washes, species-appropriate secondary antibodies with fluorescence for 488, 555, and 647 were used to visualize the immunoreaction (1:200 – 1:500, Jackson ImmunoResearch Labs donkey-anti-chicken-488, # 703-545-155; donkey-anti-rabbit-647, # 711-605-152; donkey-anti-mouse-647, # 715-605-150; Life Technologies, donkey-anti-mouse-555, # A31570; donkey-anti-rabbit-555, # A31572). Cultures were counterstained with Hoechst (Bis-benzimide, 1:500, Sigma-Aldrich, # B2261) and coverslipped with EverBrite medium (Biotium, # 23001).

Experiments were carried out to confirm gene expression of OEC subtype markers in the 8-10-week-old olfactory system of rats. Soft tissues and the jaw were removed from fixed heads that were decalcified in undiluted formic acid bone decalcifier (Immunocal, Decal Chemical Corps, Tallman, NY) for 12 hrs., washed, and then infiltrated with increasing concentrations of 5% to 20% sucrose for 24 hr. Heads in 20% sucrose were heated in a 37°C oven and then placed in a gelatin sucrose mixture for 4 hrs, embedded in warm gelatin sucrose in a plastic mold, allowed to gel, frozen on dry ice and stored at -70°C. Heads were sectioned coronally at 40 µm thick, and mounted in series on 16 glass slides. Several antibodies from each of the 4 OEC clusters were chosen to determine their distribution in olfactory bulb sections. Primary antibodies used are listed in Suppl. Table 4, together with the experimental parameters used. For most antibodies, the olfactory bulb sections first underwent a heat-induced antigen retrieval step with 10mM citric acid at pH 6 for 5 min and then were transferred to 1.5% NDS and 0.1% Triton-X-100 in TBS-BSA (0.1M Tris; 1.4% NaCl; 0.1% BSA) for 1 h followed by overnight incubation in primary antibody at RT. Following multiple washes, species-appropriate secondary antibodies with fluorescence for 488 and 555 were used to visualize the immunofluorescence.

For experiments carried out using a Tyramide Signal Amplification (TSA) kit, sections were first incubated in 1% hydrogen peroxide and 0.1% sodium azide in phosphate-buffered saline PBS for 30 mins followed by 10% NDS and 0.1% Triton X-100 in PBS for 1 hr. Next, the sections were incubated with Avidin-Biotin solution (Vector laboratories; kit #SP-2001) before an overnight incubation with the primary antibody at RT. The next day, sections were washed in PBS followed by TNT buffer (0.1M Tris-HCl; 0.15M NaCl; 0.05% Tween) and then a 1 hr incubation with biotinylated horse anti-goat IgG (1:1000; Vector laboratories; #BA-9500), biotinylated donkey anti-rabbit IgG (1:1000; Jackson Immunoresearch; #711-065-152), or biotinylated donkey anti-mouse IgG (1:1000; Jackson Immunoresearch; #715-065-150) in TSA-specific blocking buffer (TNB; 0.1M Tris-HCl; 0.15M NaCl; 0.5% Blocking reagent; PerkinElmer; #FP1020). Sections were then washed with TNT followed by a 1 h incubation with streptavidin-conjugated horseradish peroxidase (1:1000; PerkinElmer; #NEL750001EA) in TNB, and a 10-min incubation with TSA Plus Fluorescein (1:150; PerkinElmer; #NEL741001KT) or TSA Plus Cyanine 3 (Cy3; 1:100; PerkinElmer; #NEL744001KT). Sections were imaged on an Olympus AX70 microscope with a Zeiss AxioCam HRcRv.2 camera. Images were transferred to Adobe Photoshop for assembly, cropping, and adjustment of brightness/contrast to create the compiled figures.

Morphological analyses of Ki67 OEC subtypes

To determine if OEC progenitor cells marked with Ki67 immunoreactivity have a distinctive morphology, purified and fixed OEC cultures from 4 rats were processed with anti-*Ngfr*^{p75}, anti-Ki67 and counterstained with Hoechst (Bis-benzimide, 1:500, Sigma-Aldrich, #B2261). Images were acquired from 7-10 randomly selected fields/sample using an Olympus AX70 microscope and Zen image processing and analysis software (Carl Zeiss). We distinguished the larger, flat ‘astrocyte-like’ OECs from the smaller, fusiform ‘Schwann cell-like’ OECs, and recorded their expression of *Ngfr*^{p75} and Ki67. Cell counts from each field were averaged per rat and then averaged into a group mean ± SEM. A Student t-test was conducted to compare the effect of *Ngfr*^{p75}-labeled cell morphology and the proliferative marker Ki67. Statistical significance was determined by $p < 0.05$.

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Data availability

Raw scRNA-seq data and expression matrix is available at GEO (accession number GSE215247). The scRNA-seq results also can be viewed interactively at the Single Cell Portal. OEC vs Leftovers: https://singlecell.broadinstitute.org/single_cell/study/SCP1237/leftovers-and-oecs . OEC subclusters: https://singlecell.broadinstitute.org/single_cell/study/SCP1489/oec-subcluster-batch-corrected 

Conflict of Interest

The authors declare no competing interests.

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Joint Public Review:

Summary

This manuscript explores the transcriptomic identities of olfactory ensheathing cells (OECs), glial cells that support life-long axonal growth in olfactory neurons, as they relate to spinal cord injury repair. The authors show that transplantation of cultured, immunopurified rodent OECs at a spinal cord injury site can promote injury-bridging axonal regrowth. They then characterize these OECs using single-cell RNA sequencing, identifying five subtypes and proposing functional roles that include regeneration, wound healing, and cell-cell communication. They identify one progenitor OEC subpopulation and also report several other functionally relevant findings, notably, that OEC marker genes contain mixtures of

other glial cell type markers (such as for Schwann cells and astrocytes), and that these cultured OECs produce and secrete Reelin, a regrowth-promoting protein that has been disputed as a gene product of OECs.

Strengths

This manuscript offers an extensive, cell-level characterization of OECs, supporting their potential therapeutic value for spinal cord injury and suggesting potential underlying repair mechanisms. The authors use various approaches to validate their findings, providing interesting images that show the overlap between sprouting axons and transplanted OECs, and showing that OEC marker genes identified using single-cell RNA sequencing are present *in vivo*, in both olfactory bulb tissue and spinal cord after OEC transplantation.

Challenges

Despite the breadth of information presented, and although many of the suggestions in the initial review were addressed well, some points related to quantification and discussion of sex differences are not fully addressed in this revision.

(1) The request for quantification of OEC bridges is not fully addressed. We note that this revision includes the following statement (page 6): "We note, however, that such bridge formation is rare following a severe spinal cord injury in adult mammals." However, the title of the paper states that olfactory ensheathing cells promote neural repair and the abstract states that "OECs transplanted near the injury site modify the inhibitory glial scar and facilitate axon regeneration past the scar border and into the lesion." Statements such as these make it more crucial to include quantification of OEC bridges, because if single images are shown of remarkable, unusual bridges, but only one sentence acknowledges the low frequency of this occurrence, then this information taken together might present the wrong takeaway to readers.

Including some sort of quantification of bridging, whether it be the number of rats exhibiting bridges, the percentage area of OECs near a lesion site, or some other meaningful analysis, would add rigor and clarity to the manuscript.

(2) The additional discussion of sex differences in OEC bridging elaborates on the choice to study female rats, citing bladder challenges in male rats, but does not note salient clinical implications of this choice. Men account for ~80% of spinal cord injuries and likely also have worsened urinary tract issues, so it would be important to acknowledge this clinical fact and consider including males in future studies.

<https://doi.org/10.7554/eLife.95629.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Joint Public Review:

Summary

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This manuscript offers an extensive, cell-level characterization of OECs, supporting their potential therapeutic value for spinal cord injury and suggesting potential underlying repair mechanisms. The authors use various approaches to validate their findings, providing interesting images that show the overlap between sprouting axons and transplanted OECs, and showing that OEC marker genes identified using single-cell RNA sequencing are present in vivo, in both olfactory bulb tissue and spinal cord after OEC transplantation.

Despite the breadth of information presented, however, further quantification of results and explanation of experimental approaches would be needed to support some of the authors' claims. Additionally, a more thorough discussion is needed to contextualize their findings relative to previous work.

(1) a. Important quantification is lacking for the data presented. For example, multiple figures include immunohistochemistry or immunocytochemistry data (Figures 1, 5, 6), but they are presented without accompanying measures like fractions of cells labeled or comparisons against controls.

We would like to clarify that the immunohistochemistry or immunocytochemistry data presented are meant to be qualitative rather than quantitative. The main purpose of the images is to show the presence or absence of markers of OEC subtypes rather than how much is present. That being said, in the revision we now add quantitative estimates of cell fractions for OECs along with other major cell types in Supplemental Table 1 and each OEC subtype marker in Supplemental Table 2.

b. As a result, for axons projecting via OEC bridges in Figure 1, it is unclear how common these bridges are in the presence or absence of OECs.

We note that the number of spinal cord transected rats with bridges of axons crossing the lesion core are extremely rare following a severe spinal cord injury in adult mammals. Our first example of axon bridging following a complete spinal cord transection followed by OEC transplants was reported in Thornton et al., (2018) and compared to an incomplete transection in a fibroblast-transplanted control in his Figure 4. That figure also appeared the cover of Experimental Neurology when the paper was published. Figure 1 in the current paper was from an independent experiment which replicated the previously observed rare bridge formation. We noted this in the revised manuscript.

Page 6: "We note, however, that such bridge formation is rare following a severe spinal cord injury in adult mammals."

c. For Figure 6., it is unclear whether cells having an alternative OEC morphology coincide with progenitor OEC subtype marker genes to a statistically significant degree. (see top paragraph on page 11)

Franceschini & Barnett (1996) suggested that there were 2 distinct types of OECs that could be distinguished by their different morphology: one type resembling a Schwann cell and the other, an astrocyte. The purpose of Figure 6 is to determine if there is a link between our OEC subtypes based on scRNAseq with those previously described based on morphology alone (Franceschini and Barnett, 1996). There could be agreement between large, flat or small

fusiform OECs morphological and their progenitor status, but it is not required that the two classification types would significantly overlap. Here we report the percentage of morphology-based cell subtypes that show expression of our OEC subtype markers to estimate the overlap between the two. Our results indicate the two types of OEC morphologies share a certain degree of overlap, a finding that indicates similarities as well as differences between the two classification methods.

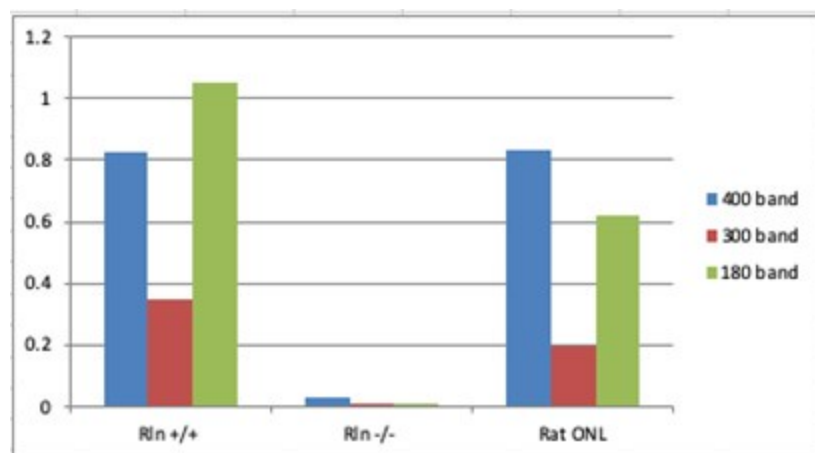
In our results section we show that $\sim 3/4$ ths of the Ki67-expressing OEC progenitor cells sampled were astrocyte-like, i.e., flat in shape and weakly Ngfr^{p75}-labeled. The remaining $\sim 1/4$ th of the Ki67-labeled OECs were fusiform in shape and expressed Ngfr^{p75} strongly. We feel that this is important to include as it is the only previous report of OB-OEC subtypes. The statistics of these results were in our original manuscript on page 11 and we further revise the text as follows:

Page 12: “To determine if the proliferative OECs differ in appearance from adult OECs, and whether there is concordance between our OEC subtypes based on gene expression markers and previously described morphology-based OEC subtyping (Franceschini & Barnett, 1996), we analyzed OECs identified with the anti-Ki67 nuclear marker and anti-Ngfr^{p75} (Figure 6g-h). Of the Ki67-positive OECs in our cultures, $24\% \pm 8\%$ were strongly Ngfr^{p75}-positive and spindle-shaped, whereas $76\% \pm 8\%$ were flat and weakly Ngfr^{p75}-labeled (n=4 cultures, $p=0.023$). Here we show that a large percentage ($\sim 3/4$ ths) of proliferative OECs are characterized by large, flat morphology and weak Ngfr^{p75} expression resembling the previously described morphology-based astrocyte-like subtype. Our results indicate the two types of OEC classifications share a certain degree of overlap, indicating similarities but also differences between the two classification methods.”

d. Similar quantification is missing in other types of data such as Western blot images (Fig. 9) and OEC marker gene data (for which p-values are not reported; Table S2).

Response on Western blots: The Western blot signals shown in Figure 9 are from experiments that were designed to be qualitative rather than quantitative, by addressing the question, “Can we detect Reelin signals or not? in the different samples.” Both Western blots show that *Reln*^{+/+} mouse olfactory bulbs (d) or cortices (e) contain Reelin whereas *Reln*^{-/-} samples do not and therefore provide positive and negative controls, respectively. The rat olfactory nerve layer (ONL, laminae I-II of olfactory bulb, d lane 1; e lane 3) contains mainly OECs wrapped around the axons of the olfactory sensory neurons that transmit olfactory signals into the olfactory bulb. To address your request for quantification, Dr. Khankan measured the density of the three isoforms of Reelin, 400 kD, 300 kD and 180 kD in Fig. 9e and normalized them against the GAPDH control (37 kD). The graph below shows the normalized band density in arbitrary units on the Y-axis relative to the first 3 conditions, i.e., *Reln*^{+/+} and *Reln*^{-/-} mouse cerebral cortices and rat *Reln*^{+/+} ONL. Because the conditioned medium was collected from tissue culture medium rather than cells or tissue, the GAPDH control was not present and therefore these data cannot be normalized in a similar analysis.

Author response image 1.



Response for OEC marker gene data: We now add new full supplementary Table S1 (for major cell types) and Table S2 (for OEC subtypes) to report statistical p values and adjusted p values, as well as additional statistics information including percent cell expressing a subtype marker in a given subtype versus in other subtypes.

e. The addition of quantitative measures and, where appropriate, statistical comparisons with p-values or other significance measures, would be important for supporting the authors' claims and more rigorously conveying the results.

As detailed in the above responses, we now add quantifications and statistics to support the claims and enhance the rigor of our analysis.

(2) a. Some aspects of the experimental design that are relevant to the interpretation of the results are not explained. For example, OECs appear to be collected from only female rats, but the potential implications of this factor are not discussed.

We added a short explanation in the Discussion and Methods section regarding why spinal cord injury studies are carried out on female rats.

Page 24, Discussion: “Due to the extensive urinary tract dysfunction in spinal cord transected rats, most studies prefer females as their short urethra facilitates daily manual bladder expression. Our study, therefore, was carried out only on adult female rats, so sex differences and the generalizability of our findings to adult male rats would require further investigation.”

Page 26, Methods: “Only females were used in order to match the sex of previous SCI studies conducted exclusively on female rats (Dixie, 2019; Khankan et al., 2016; Takeoka et al., 2011; Thornton et al., 2018). Following complete thoracic spinal cord transection, an adult rat is unable to urinate voluntarily and therefore urine must be manually “expressed” twice a day throughout the experiment. Females have a shorter urethra than males, and thus their bladders are easier to empty completely.”

b. Additionally, it is unclear from the manuscript to what degree immunopurified cells are OECs as opposed to other cell types. The antibody used to retain OECs, nerve growth factor receptor p75 (Ngfr-p75), can also be expressed by non-OEC olfactory bulb cell

types including astrocytes [1-3]. The possible inclusion of Ngfr-p75-positive but non-OEC cell types in the OEC culture is not sufficiently addressed.

(a) Cragnolini, A.B. et al., Glia, (2009), doi: 10.1002/glia.20857.

(b) Vickland H. et al., Brain Res., (1991), doi: 10.1016/0006-8993(91)91659-O.

(c) Ung K. et al., Nat Commun., (2021), doi: 10.1038/s41467-021-25444-3.

Our OECs are dissected primarily from the olfactory nerve layer that is concentrated medially and ventrally around the olfactory bulb together with a small part of the glomerular layer (layer II). OECs are the only glia present in olfactory nerve layer. Thus, although it is possible that other cell types also express Ngfr-p75 as pointed out by the reviewer and in the references provided, our OEC dissection method severely limits the number of astrocytes that might be included in our cultures. We further provide additional evidence (see updated Figure 2d and the detailed responses to the next question) that our immunopanned OECs using our dissection method consistently express all classic OEC markers but do not consistently express the majority of classic markers for other glial cell types such as astrocytes or oligodendrocytes.

Such non-OEC cell types are also not distinguished in the analysis of single-cell RNA sequencing data (only microglia, fibroblasts, and OECs are identified; Figure 2). Thus, it is currently unclear whether results related to the OEC subtype may have been impacted by these experimental factors.

We need to clarify that when determining potential cell types in Figure 2, we compared our cell cluster marker genes against a broad array of cell types including astrocytes, oligodendrocytes and Schwann cells, but the gene overlap was only significant for microglia, fibroblasts, and OECs, which we labeled in new Figure 2d. We added more details in methods and results to clarify how we determined the cell types in Figure 2 (text added below). We did consider all the potential cell types that could have been present in our OEC cultures, including astrocytes. However, astrocyte or oligodendrocyte markers were not significantly enriched in the clusters, but markers for microglia, fibroblasts, and OECs were prominent in the cell clusters.

In the revised Figure 2d, we now illustrate that the OEC clusters not only express typical OEC markers, but also express a few but not all marker genes from other glial cells. We show the comparative data on markers for astrocytes, oligodendrocytes, and Schwann cells in Figure 2d in parallel with the marker genes for OECs, microglia, and fibroblasts. For each of the other glial cell types, there are some genes which overlap with OECs, and that is the reason why we identified OECs as hybrid glia.

Page 6, Results: “Based on previously reported cell type marker genes for fibroblasts and major glial cell types including OECs, astrocytes, oligodendrocytes, and microglia, we found elevated expression of OEC marker genes in clusters 2, 3 and 7, microglia marker genes in clusters 4, 6, and 7, and fibroblast marker genes in clusters 0, 1, and 5 (Figure 2d).”

Page 33, Methods: “Additional marker genes for fibroblasts and multiple glial cell types including astrocytes, oligodendrocytes, and microglia were also used to compare with those of the cell clusters.”

(3) The introduction, while well written, does not discuss studies showing no significant effect of OEC implantation after spinal cord injury. The discussion also fails to sufficiently acknowledge this variability in the efficacy of OEC implantation. This omission amplifies bias in the text, suggesting that OECs have significant effects that are not fully reflected in the literature. The introduction would need to be expanded to properly address the

nuance suggested by the literature regarding the benefits of OECs after spinal cord injury. Additionally, in the discussion, relating the current study to previous work would help clarify how varying observations may relate to experimental or biological factors.

We appreciate the insightful comment and have now included information about the variability in OEC transplantation in previous studies in both the introduction and discussion sections. We discuss technical differences that lead to variability in the Introduction and how our findings could help interpret the variability in the Discussion.

Page 4-5: Text added to the Introduction: “The outcomes of OEC transplantation studies after spinal cord injury vary substantially in the literature due to many technical differences between their experimental designs. The source of OECs has a great impact on the outcome, with OB-OECs showing more promise than peripheral lamina propria-derived OECs, and purified, freshly-prepared OECs being required for optimal OEC survival. Other important variables include the severity of the injury (hemisection to complete spinal cord transection), the age of the spinal cord injured host (early postnatal versus adult), and OEC transplant strategies (delayed or acute transplantation, cell transplants with or without a matrix; Franssen et al., 2007). Franssen et al. (2007) evaluated studies that used only OECs as a transplant, and reported that 41 out of 56 studies showed positive effects, such as OEC stimulation of regeneration, positive interactions with the glial scar and remyelination of axons. More recent systematic reviews and meta-analyses on the effects of OEC transplantation following different spinal cord injury models reported that OECs significantly improved locomotor function (Watzlawick et al. 2016; Nakjavan-Shahraki et al., 2018), but did not improve neuropathic pain (Nakjavan-Shahraki et al., 2018.)”

Pages 24-25: Discussion on OEC source variability “Extensive differences between OEC preparations contribute to the large variation in results from OEC treatments following spinal cord injury. This scRNA-seq study focused entirely on OB-OECs, and the next step would be to carry out similar studies on the peripheral, lamina-propria-derived OECs to discern the differences between these OEC populations. Such comparative studies using scRNA-seq will help define the underlying mechanisms and help resolve the variability in results from OEC-based therapy. Detailed studies of the composition of different OEC transplant types will contribute to identifying the most reparative cell transplantation treatments.”

Reviewer #1 (Recommendations For The Authors):

This is an extremely well-written and impactful series of experiments from a renowned leader in the field. The experimental questions are timely, with similar therapeutic approaches being prepared for clinical trial. The results address a gap that has persisted in the field for several decades and one that has been considered by many scientists long before technology existed to find answers. This highlights the importance of these experiments and the results reported here. With these things in mind, there are only a few minor factors that I have, that should be addressed to strengthen the paper.

We truly appreciate the positive evaluations from the reviewer!

Primary concerns

(1) Quantification of results: The authors report on the data with broad brush strokes, missing the opportunity to quantify results and strengthen the interpretations. For instance, when describing gene expression, what proportion of cells analyzed were expressing these genes? How did this compare with detectable levels of protein? Can the author draw correlations between data sets collected that could offer even more insight into the identities of the cells studied? There is also a missed opportunity to evaluate how

transplantation into injured neural tissue might alter gene expression of the phenotypes identified prior to transplantation.

We appreciate these insightful comments and have added quantitative information and other relevant discussions in the revision. We now add Suppl Tables 1 (for major cell types including OECs, fibroblast, and microglia) and 2 (for OEC subtypes) to indicate the proportion of cells expressing each marker gene in each given cell cluster/subtype in the column. “Percentage of cells expressing the gene in the subtype/cell type” versus the proportion of cells expression the given marker genes in other cell types in the column “Percentage of cells expressing the gene in the other subtypes/cell types.” In the new supplementary tables, we report statistical p values and adjusted p values after multiple testing correction to indicate statistical significance.

Regarding the comparison with protein levels, we carried out immunohistochemistry experiments to confirm the proteins corresponding to OEC subtype markers. Our findings show that proteins for the gene markers can be detected, and thereby supports our sc-seq findings. However, the immunofluorescence only provides a qualitative measure of protein levels in situ, so we cannot perform a correlation analysis. This is something we plan to pursue in a follow-up study with measurable protein levels. We also discuss future directions to examine the genes and proteins in *in vivo* transplantation studies in the Discussion.

(2) Discussion and interpretation: Greater depth to interpretation and discussion of data and its impact on future work is needed. For example, on pages 20-21, the authors reflect briefly on why Reelin might be of interest (it could lead to Dab-1 expression), but why is that important? There are several instances like this where it would be useful for the authors to provide a little more insight into the potential impact of these data and interpretations.

We appreciate these valuable suggestions. We have revised our Results and Discussion sections to offer deeper insight and interpretation of the importance of the data, especially that for Reelin.

Page 17: Results: “In the canonical Reelin-signaling pathway, Reelin binds to the very-low-density lipoprotein receptor (Vldlr) and apolipoprotein E receptor 2 (ApoER2) and induces Src-mediated tyrosine phosphorylation of the intracellular adaptor protein Disabled-1 (Dab1). Both Reelin and Dab1 are highly expressed in embryos and contribute to correct neuronal positioning.”

Page 22-23, Discussion: “Reelin is a developmentally expressed protein detected in specific neurons, in addition to OECs and Schwann cells. The canonical Reelin-signaling pathway involves neuronal-secreted Reelin binding to Vldlr and ApoER2 receptors expressed on Dab1-labeled neurons. Following Reelin binding, Dab1 is phosphorylated by Src family kinases which initiates multiple downstream pathways. Very little is known, however, about Reelin secreted by glia. Panteri et al. (2006) reported that Schwann cells express low levels of Reelin in adults, and that it is upregulated following a peripheral nerve crush, as is reported above for many neurotrophic factors. Reelin loss in Schwann cells reduced the diameter of small myelinated axons but did not affect unmyelinated axons (Panteri et al., 2005). In the olfactory system, OECs ensheath the Dab1-labeled, unmyelinated axons of olfactory sensory neurons which are continuously generated and die throughout life. OEC transplantation following spinal cord injury would provide an exogenous source of Reelin that could phosphorylate Dab1-containing neurons or their axons. Dab1 is expressed at high levels in the axons of some projection neurons, such as the corticospinal pathway (Abadesco et al., 2014). Future experiments are needed to explore the function that glial-secreted Reelin may have on axonal regeneration.”

(3) The authors reflect on the spontaneous glial bridge that develops in the repairing spinal cord of Zebrafish, but perhaps even more relevant is that this same phenomenon occurs in mammals as well if the spinal cord is injured during early development (opossum; Lane et al, EFN 2007). This should be considered and the statement that there is little regeneration in the mammalian spinal cord should be clarified.

We appreciate this insightful comment. We now add discussions of the axonal regeneration and bridging observed following severe spinal cord injury in young developing mouse and opossum spinal cords.

Page 23: “Adult mammals show little evidence of spontaneous axonal regeneration after a severe spinal cord injury in contrast to transected neonatal rats (Bregman, 1987; Bregman et al., 1993) and young postnatal opossums (Lane et al., 2007). In immature mammals, axons continue to project across or bridge the spinal cord transection site during development. Lower organisms such as fish, show even more evidence of regeneration following severe SCI. Mokalled et al. (2016) reported that glial secretion of Ctgfa/Ccn2 was both necessary and sufficient to stimulate a glial bridge for axon regeneration across the zebrafish transection site. Cells in the injury site that express Ctgf include ependymal cells, endothelial cells, and reactive astrocytes (Conrad et al., 2005; Mokalled et al., 2016; Schwab et al., 2001). Here we show that, although rare, Ctgf-positive OECs can contribute to glial bridge formation in adult rats. The most consistent finding among our severe SCI studies combined with OEC transplantation is the extent of remodeling of the injury site and axons growing into the inhibitory lesion site, together with OECs and astrocytes. The formation of a glial bridge across the injury was critical to the spontaneous axon generation seen in zebrafish (Mokalled et al., 2016) and likely contributed to the axon regeneration detected in our OEC transplanted, transected rats (Dixie, 2019; Khankan et al., 2016; Takeoka et al., 2011; Thornton et al., 2018).

Reviewer #2 (Recommendations For The Authors):

(1) The manuscript title and abstract must include the species and sex studied.

The title and abstract have been modified as suggested.

Page 1: “Olfactory ensheathing cells from adult female rats are hybrid glia that promote neural repair”

(2) OECs submitted for sequencing were like those about to be transplanted; however, the phenotype of the cells would likely change immediately and shift over time post-implantation. Please briefly address or discuss this point in the Discussion (or Results).

We have added this important discussion point.

Pages 23-24: Discussion: “We recognize that this study is a single snapshot of OEC gene expression derived from adult female rats before they are transplanted above and below the spinal cord transection site. We would expect the gene expression of transplanted OECs to change in each new environment, i.e. as they migrate into the injury site, integrate into the glial scar, and wrap around axons. Based on our past studies, OECs survived in an outbred Sprague-Dawley rat model for ~ 4 weeks (Khankan et al., 2016) and in an inbred Fischer 344 model for 5 months (Dixie, 2019). As spinal cord injury transplant procedures are further enhanced and OEC survival improves, these hybrid glial cells should be examined at multiple time points to better evaluate their proregenerative characteristics.”

(3) Page 12: Use of "monocytes" - the word "monocyte" implies a circulating, undifferentiated innate immune cell. This should not be used interchangeably with macrophage or microglia.

We agree and now refer to microglia or macrophages depending on the context. We did leave the term monocyte in Table 3 if these cells were found in a top 20 gene reported in the references.

(4) Page 12: "We now show that these unique monocytes reported between the bundles of olfactory axons surrounded by OECs (Smithson & Kawaja, 2010), are in fact, a distinct subtype of OECs."

Is it possible to conclude that these cells are a "distinct subtype of OECs?" Perhaps these cells are a hybrid between microglia/macrophages and OECs? This is speculative, so should be worded more carefully - especially in the Results section. Please clarify, dampen conclusions, and/or better justify the wording here.

We agree and have modified the entire paragraph to dampen and more carefully explain our conclusions. We also added an additional observation that strengthens the relationship between OECs and microglial/macrophages.

Page 12, Results: Additional observation: "In fact, all top 20 genes in cluster 3 are expressed in microglia, macrophages, and/or monocytes (Suppl. Table 3)."

Page 13, Results: The statement referenced in your review was deleted and we wrote the following: "Smithson and Kawaja (2010) identified unique microglial/macrophages that immunolabeled with Iba-1 (*Aif1*) and Annexin A3 (*Anxa3*) in the olfactory nerve and outer nerve layer of the olfactory bulb. These authors proposed that Iba1-Anxa3 double-labeled cells were a distinct population of microglia/macrophages that protected the olfactory system against viral invasion into the cranial cavity. Based on our scRNA-seq data we offer an alternative interpretation that at least some of these Iba-1-Anxa3 cells may be a hybrid OEC-microglial cell type. Supporting this interpretation, there are a number of reports that suggest OECs frequently function as phagocytes (e.g., Khankan et al., 2016; Nazareth et al., 2020; Su et al. 2013)."

(5) Page 13: "Pseudotime trajectory analysis, a widely used approach to predict cell plasticity and lineages based on scRNA-seq data, suggests that there are potential transitions between specific OEC subclusters." This is interesting but is somewhat unclear. Please add one more sentence to aid the reader's understanding regarding how this analysis is performed.

Thank you for your valuable feedback. We have revised the text for clarity as follows:

Page 14, Results: "We performed pseudotime trajectory analysis using the Slingshot algorithm to infer lineage trajectories, cell plasticity and lineages by ordering cells in pseudotime based on their transcriptional progression reflected in scRNA-seq data. Transcriptional progression refers to the changes in gene expression profiles of cells as they undergo differentiation or transition through different states. The trajectory analysis results suggest that there are potential transitions between specific OEC subclusters."

(6) The authors could discuss potential reasons for variability in OEC treatment results after spinal cord injury between studies and labs. How might sequencing results here inform the debate about whether OECs are helpful or not?

In response to the Public Review, we added discussions about the variability in OEC treatments between studies in both the Introduction and Discussion, and these comments are copied on pages 6-7 of this document. In the Discussion we included a statement about how the current findings may inform the debate on OECs.

(7) Discussion: please add a discussion of limitations and future directions that addresses the following points:

a) Please add one sentence on the lack of studying sex differences - only females were studied here.

b) There is no correlation or modulation of any target genes, so all results here are correlative.

c) Please add a brief paragraph with future directions for the field, including acknowledgment that the role of OECs in repair after SCI is not fully resolved and that future studies might consider targeting some of the specific pathways described herein.

d) Which pathways and OEC subpopulations likely best support repair, and how might these be reinforced or better maintained in the SCI environment? If not known, what are the next steps for identifying the most reparative OEC subtype?

Thank you for the valuable suggestions. We have added these to the discussion as detailed below.

Pages 23-25, Discussion:

“Limitations of these OEC scRNA-Seq studies”

“We recognize that this study is a single snapshot of OEC gene expression derived from adult female rats before they are transplanted above and below the spinal cord transection site. We would expect the gene expression of transplanted OECs to change in each new environment, i.e. as they migrate into the injury site, integrate into the glial scar, and wrap around axons. Based on our past studies, OECs survived in an outbred Sprague-Dawley rat model for ~ 4 weeks (Khankan et al., 2016) and in an inbred Fischer 344 model for 5 months (Dixie, 2019). As spinal cord injury transplant procedures are further enhanced and OEC survival improves, these hybrid glial cells should be examined at multiple time points to better evaluate their proregenerative characteristics.”

“Due to the extensive urinary tract dysfunction in spinal cord transected rats, most studies are conducted on females as their short urethra facilitates daily manual bladder expression. Our study was carried out only on adult female rats, so sex differences and the generalizability of our findings to adult male rats would require further investigation. We also did not modulate any of the genes or proteins in the identified OEC subtypes to test their causal and functional roles, thus our findings remain correlative in the current study. Future gene/protein modulation studies are necessary to understand the functional roles of the individual OEC subtypes in the context of their reparative functions to determine which pathways and subtypes are more critical and can be enhanced for neural repair. Our current findings build the foundation for these future studies to help resolve the role of OECs in spinal cord injury repair.”

“Extensive differences between OEC preparations contribute to the large variation in results from OEC treatments following spinal cord injury. This scRNA-seq study focused entirely on OB-OECs, and the next step would be to carry out similar studies on the peripheral, lamina-propria-derived OECs to discern the differences between the two OEC populations. Such comparative studies using scRNA-seq will help define the underlying mechanisms and resolve the variability in results from OEC-based therapy. Detailed studies of the composition of different OEC transplant types will contribute to identifying the most reparative cell transplantation treatments.”

(8) Figure 6: What is the major point of this figure and its related immunocytochemistry? Please clarify.

Franceschini & Barnett (1996) suggested that there were 2 distinct types of OECs that could be distinguished by their different morphology: One type resembling a Schwann cell and the other, an astrocyte. The purpose of Figure 6 is to determine if there is a link between our scRNA-seq-based OEC subtypes with those previously described based on morphology alone (Franceschini and Barnett, 1996). In our results section we show that ~3/4ths of the OECs sampled that were Ki67+ progenitor cells and were astrocyte-like, i.e., flat in shape and weakly Ngfr^{p75}-labeled. The remainder were Schwann cell-like, fusiform in shape and strongly Ngfr^{p75}-labeled. Our results indicate the two types of OEC classifications share certain degrees of overlap, indicating similarities but also differences between the different classification methods.

(9) Figure 9, caption: "OEC whole cell lysates (WCL; lanes: 4, 6, and 8), and OEC conditioned medium (CM; lanes: 5 and 7)." This statement is unclear - please clarify the result here.

We added clarification to the legend for Figure 9d.

Page 50: (d) "Western blot confirms the expression of Reelin in rat olfactory nerve layer I and layer II (ONL; lane 1 of western blot). *Reln*^{+/+} and *Reln*^{-/-} mouse olfactory bulbs were used as positive and negative controls, respectively (lanes: 2 and 3). Reelin that was synthesized by cultured OECs was found in whole cell lysates (WCL; lanes: 4, 6, and 8), whereas Reelin that was secreted by cultured OECs into tissue culture medium was measured in the OEC "conditioned medium" (CM; lanes: 5 and 7). GAPDH was the loading control for tissue homogenates (lanes 1-4, 6, 8)."

(10) Methods: A Cat. No. for all antibodies and key supplies should be included.

Response: All of the antibody information in the revised version is in Suppl. Table 4. Information for other key supplies is included in the extensive methods section.

(11) Methods: How was primary antibody specificity validated for less-used antibodies? Background staining can be a major issue after SCI; e.g., with the CTGF antibody used in Figure 5.

The spinal cord section shown in Figure 5 was compared to sections from the same SCI cohort that had been injected with control cells, i.e. skin fibroblasts. We have used the first two antibodies (anti-Glial fibrillary acidic protein and anti-Green fluorescent protein) for many years so only the CTGF was a "less-used antibody." Our strategy for working with "less-used" or "newly-purchased" antibodies was as follows.

First, we studied the literature to find the best antibodies for neuronal tissue. Many of the images in Figure 7 were generated with antibodies purchased just for this study. Our goal was to characterize them on normal adult lamina propria and olfactory bulb tissues rather than in the injured spinal cord where background can be an issue. In the olfactory bulb we examined the olfactory nerve layer where OECs are concentrated and then examined the olfactory epithelium, lamina propria, and the deep layers of the olfactory bulb to find regions without immunolabel. As described above, we tested anti-CTGF antibodies in SCI sections implanted with skin fibroblasts controls when conducting experiments for CTGF in sections with OECs. New antibodies were tested at multiple concentrations and we tried different immunocytochemical techniques. Anti-CTGF is expressed by several different cell types, but expression is low in most of the areas above and below the injury site. Despite our success

with many “newly-purchased” antibodies there were at least 4 of them that we were never able obtain specific labeling.

| (12) *Will the data (especially the sequencing data) be shared publicly?*

The data has been uploaded to and shared via the public data repository GEO. Data availability is stated on the title page of this manuscript.

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